Splunk4Ninjas - Data Onboarding

Lab Guide

Overview

This lab guide contains the step by step instructions for hands-on exercises done in the Splunk4Ninjas - Data Onboarding workshop. This workshop simulates the following:

- Monitoring a location on either your local Splunk instance or from a Splunk Universal/Heavy Forwarder
- Getting data in from a data source that may not have a Splunk app or add-on currently available

Please ensure that you have a copy of the workshop slide deck: https://splk.it/S4N-DataOnboarding

Prerequisites

Splunk Instance

In order to take part in the workshop exercises, you will need your own Splunk instance. You will either be provided an instance by your workshop host or you will need to enroll in the workshop to have an instance provisioned. Your host will let you know.

If you are instructed to enroll in today's workshop, you will need to have a Splunk.com account prior to enrolling. If you don't already have a Splunk.com account, please create one at https://www.splunk.com/en_us/sign-up.html before proceeding with the enrollment link provided by your host.

Knowledge Requirements

Attendees should have a base knowledge of the Splunk interface and understand key concepts such as:

- What is an index?
- What is a sourcetype?
- What is a data model?

⚠ Troubleshooting Connectivity

If you experience connectivity issues with accessing either your workshop environment or the event page, please try the following troubleshooting steps. If you still experience issues please reach out to the team running your workshop.

- **Use Google Chrome** (if you're not already)

- If the event page (i.e. *https://show.splunk.com/event/<eventID>*) didn't load when you clicked on the link, try **refreshing the page**
- **Disconnect from VPN** (if you're using one)
- **Clear your browser cache and restart your browser** (if using Google Chrome, go to: Settings > Privacy and security > Clear browsing data)
- **Try using private browsing mode** (e.g. Incognito in Google Chrome) to rule out any cache issues
- **Try using another computer** such as your personal computer - all you need is a web browser! Cloud platforms like AWS can often be blocked on corporate laptops.

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Lab 0 - Register/Enroll and Login to Instance

Description

You will need a Splunk training instance to take part in the labs for the workshop. In this lab, you will register for your own Splunk instance to use for the workshop.

🔑 **Has your Host already provided you a link for your instance and login credentials?**

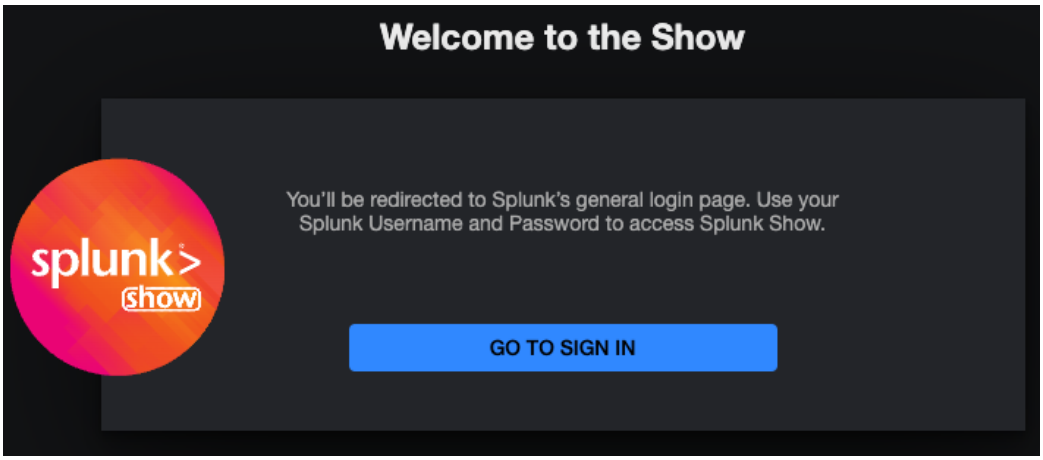
If your workshop host has already provided you with an instance link as well as login credentials, **you do NOT need to follow these instructions for Lab 0**. You can instead skip straight to Lab 1 - File Monitor!

Steps

Instance Creation and Logon

---

1. Browse to https://show.splunk.com - or the enrollment link provided by the host - and log in using your Splunk.com account credentials:



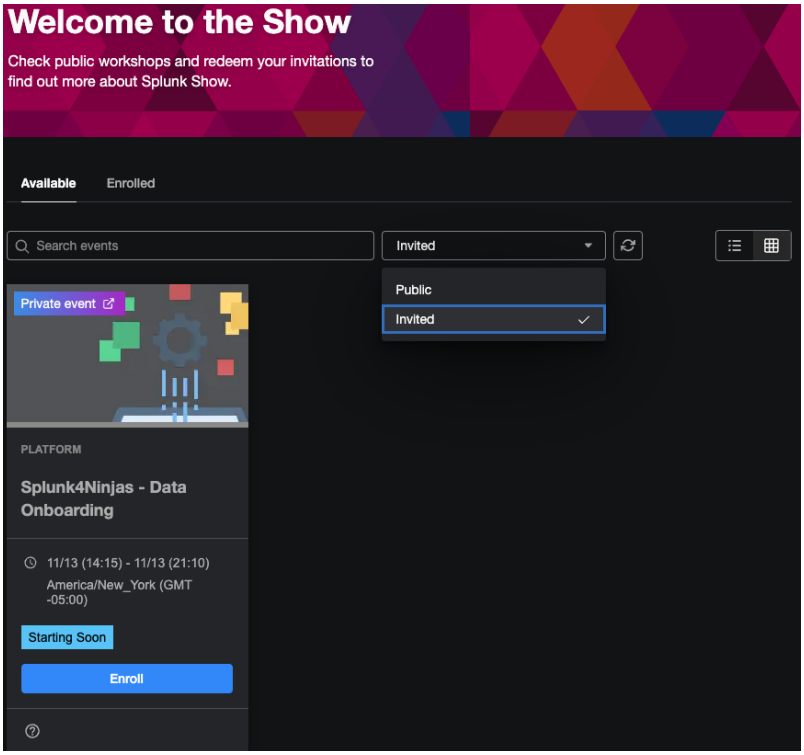
Don't have a Splunk.com account?

If you do not have a Splunk.com account, create one in a few minutes by clicking here. After creating your Splunk.com account, please navigate back to https://show.splunk.com or use the enrollment link your host provides.

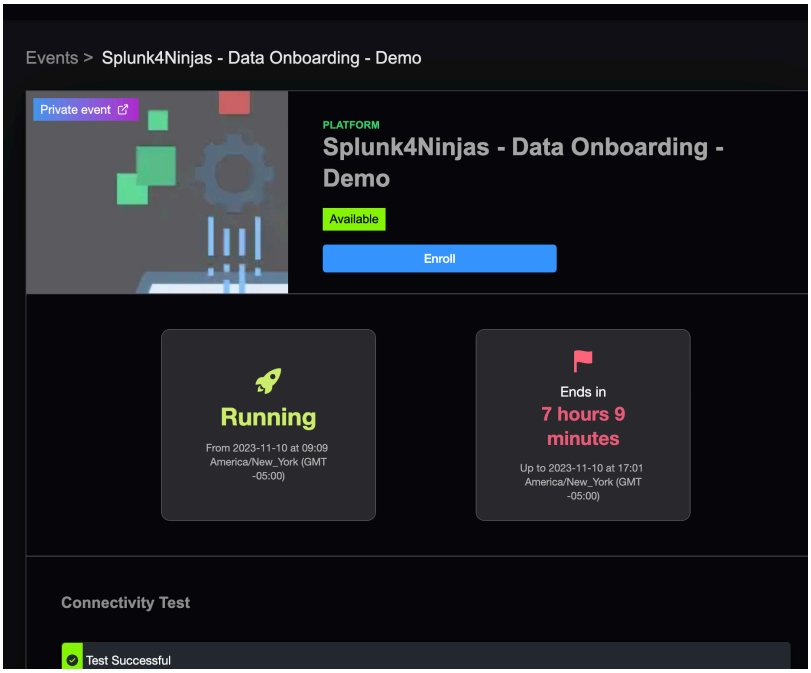
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2. Once logged into Splunk Show, you will see the event page for the event that you have been invited to. If nothing is showing, try selecting “Invited” from the dropdown list.

   (View if you go directly to https://show.splunk.com and login)



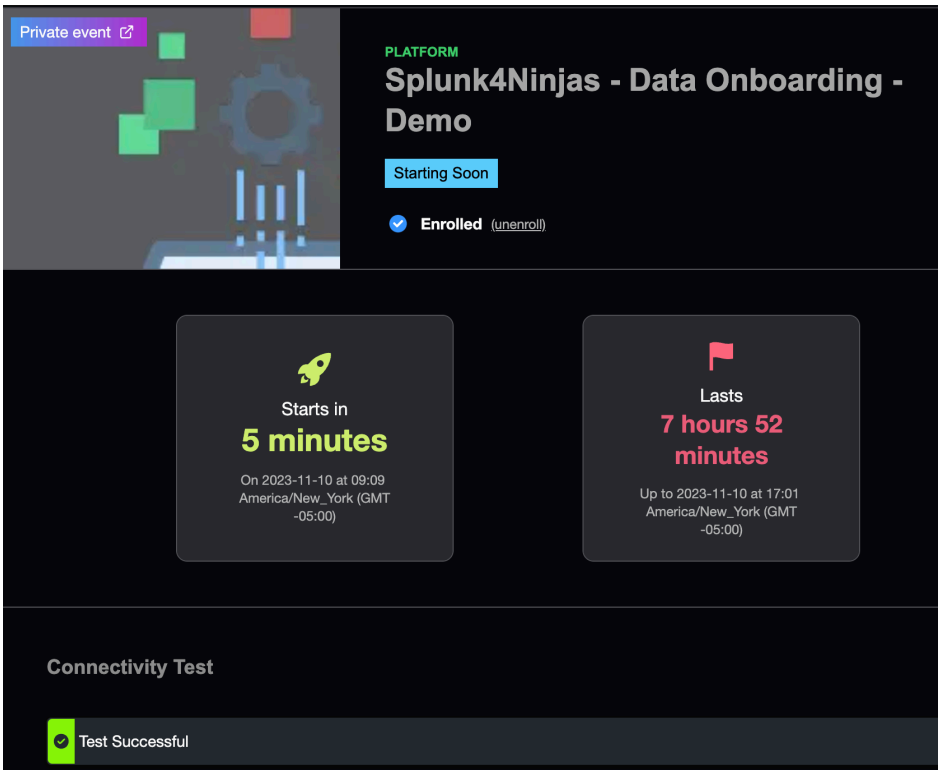
   (View from the enrollment link directly)



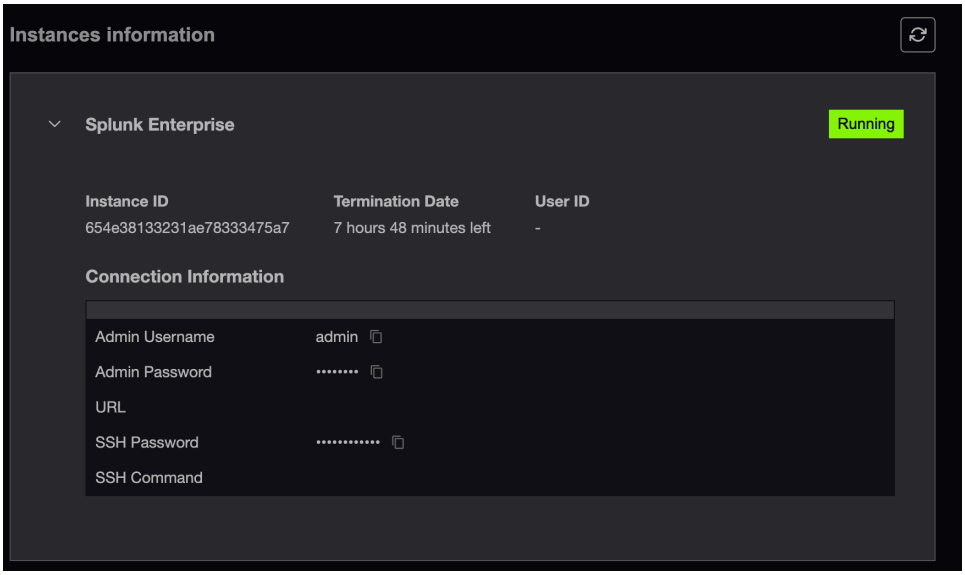
3. Click on **Enroll** to join the event.

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4. The page will refresh and the event will now display ‘Enrolled’



5. Once the workshop starts, your individual environment will start up.
6. Scroll down the page to the **Instances Information** section and expand the **Splunk Enterprise** section to locate the URL and login credentials for your lab environment.



If you don’t see any connection information displayed yet, it means that your lab environment has not yet fully provisioned. Please check this page again in a few minutes.

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🕑 **Lab environment expiration**

All Splunk instances that are part of this workshop will automatically be shut down at the termination date specified under your instance information so feel free to continue to explore your lab environment until then!

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Lab 1 - File Monitor

Description

Firewall logs collected by a Syslog server are stored on the server locally. We will bring those firewall logs into Splunk by monitoring the files and directories where those logs are stored. With Splunk Enterprise, we can do this from either the Splunk Web interface (GUI) or the inputs.conf file through the CLI:

- Option 1 - Monitoring through the GUI
- Option 2 - Monitoring through the CLI

To configure an input, a stanza needs to be added to the inputs.conf file in the **$SPLUNK_HOME/etc/system/local/** directory or an application directory in **$SPLUNK_HOME/etc/apps/<app_name>/local/**. The directory paths are locations on the machine that runs Splunk Enterprise or the Universal Forwarder. Settings we adjust in the GUI will be reflected in those underlying files.

---

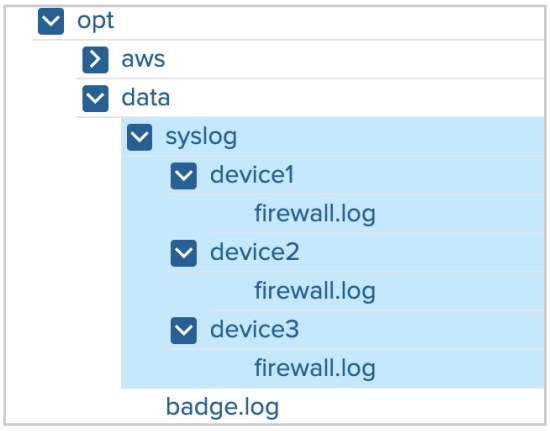
inputs.conf

To learn more about the inputs.conf file, click here.

Option 1: Monitoring Through the GUI

Monitor the Syslog Directory

1. Select Source by navigating to **Settings > Add Data > Monitor > Files & Directories** and clicking on **Browse**.
2. Select the **opt > data > syslog** directory. Ensure all devices and firewall.log files are highlighted and click the **Select** button in the bottom right of the pop up.

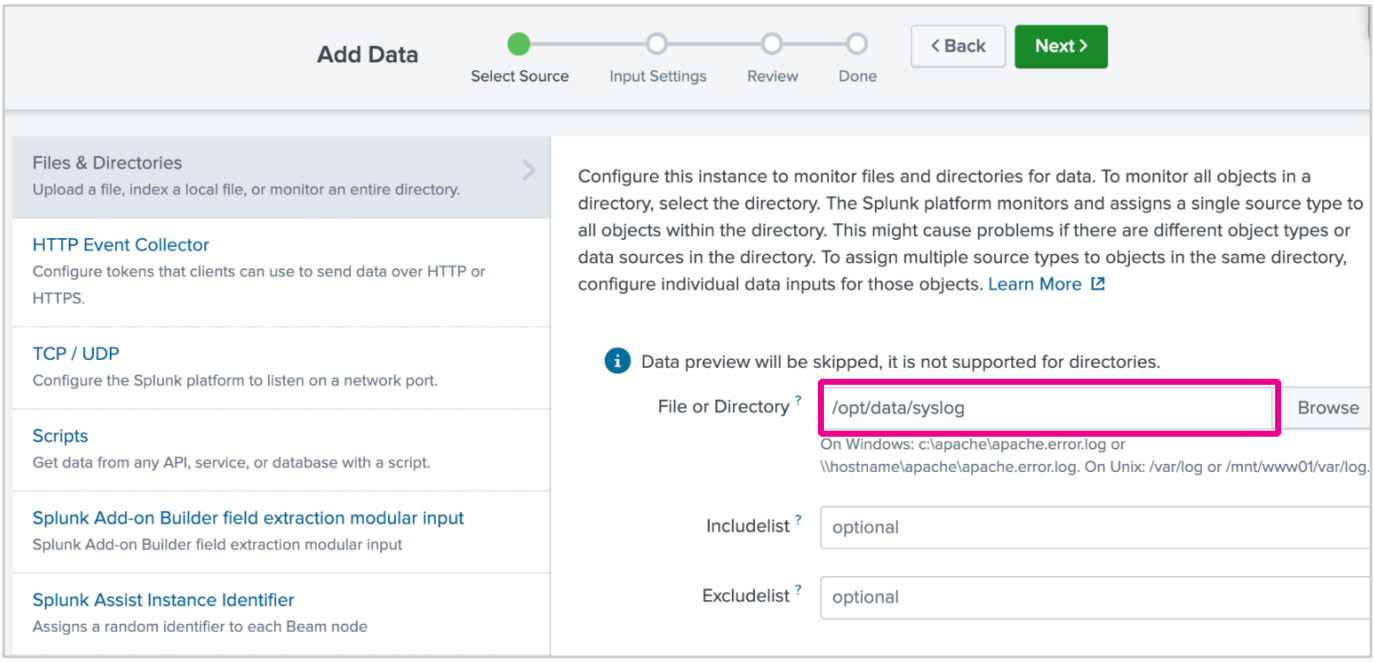


3. Back on the **Select Source** page, your **File or Directory** field should read:

/opt/data/syslog

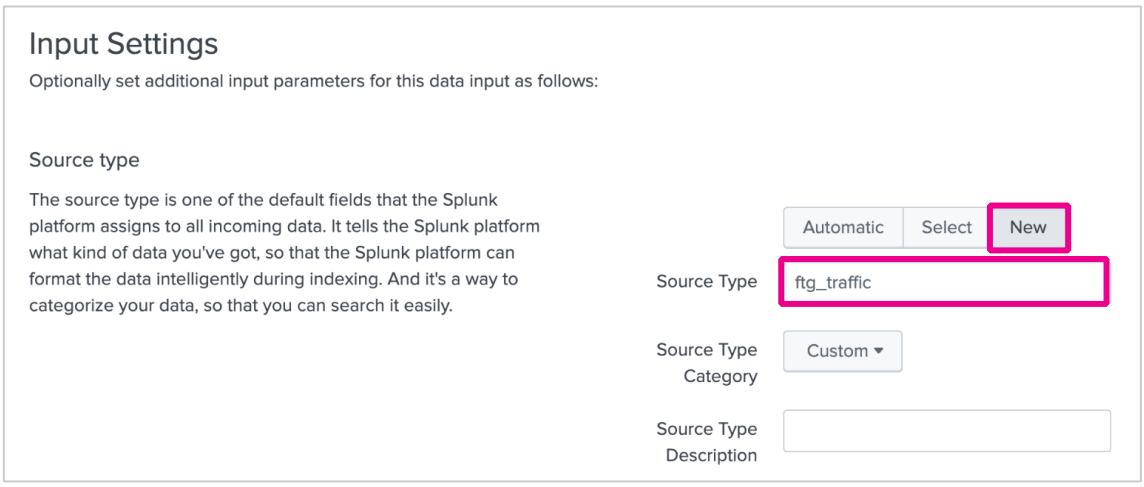
With this set, click **Next**.

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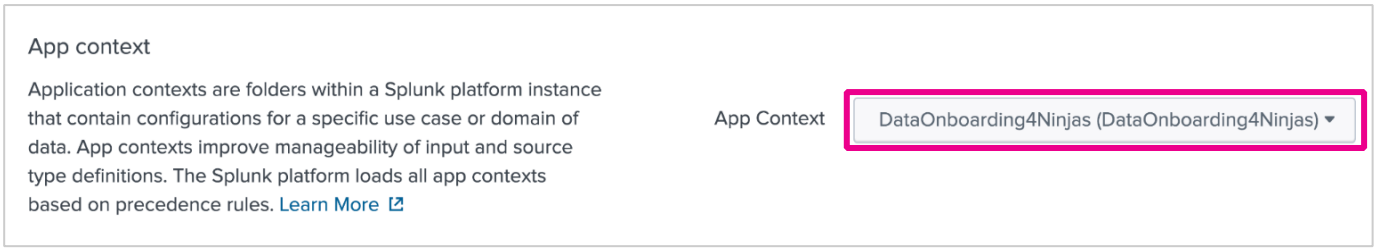


Adjust the Input Settings

4. On the **Input Settings** page, set the Source type by selecting **New** and entering **fgt_traffic** for the **Source Type** field:

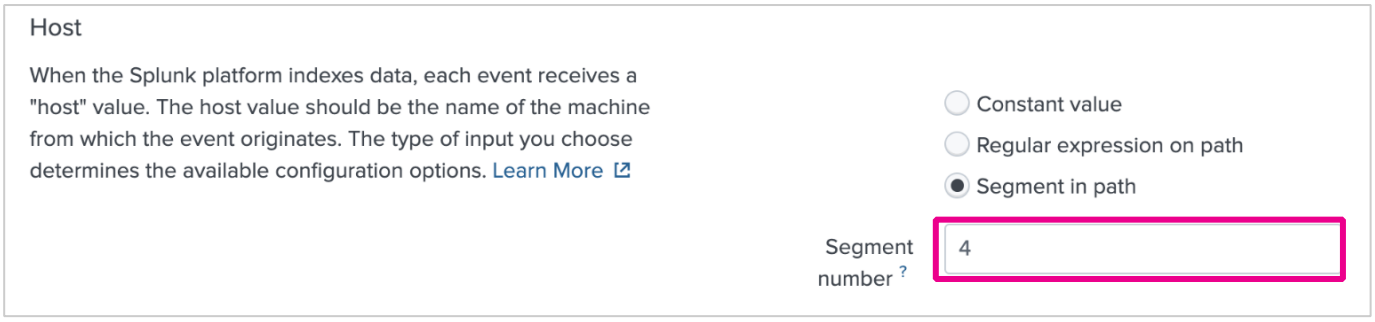


5. Set the **App Context** to **DataOnboarding4Ninjas**:



6. For the **Host** select **Segment in path** and set the **Segment number** to **4**:

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ℹ️ Segment in path

These settings specify that the “host” value assigned to the events will align with whatever comes after the 4th slash in the directory path. We are monitoring firewall logs from the following directories:

/opt/data/syslog/device1

/opt/data/syslog/device2

/opt/data/syslog/device3

which means that the device# is the 4th segment of the file path and will get assigned as the host depending on where the logs come from, i.e:

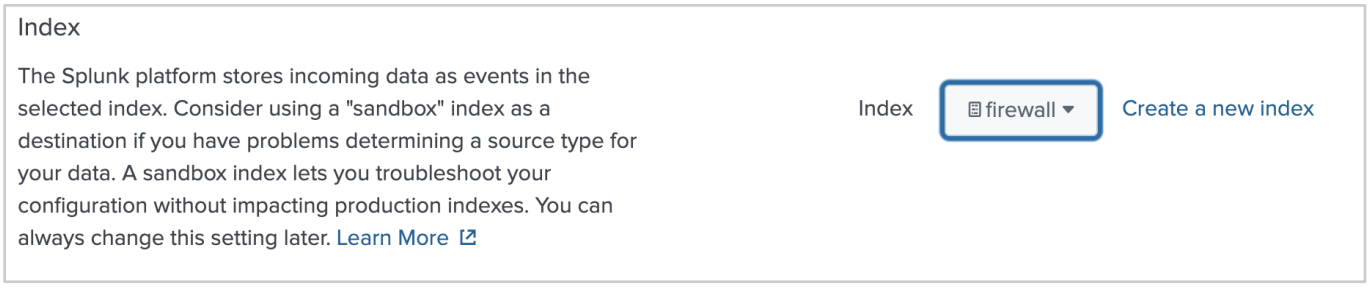
/opt = segment 1

/data = segment 2

/syslog = segment 3

/device* = segment 4

7. Set the **Index** to **firewall**:

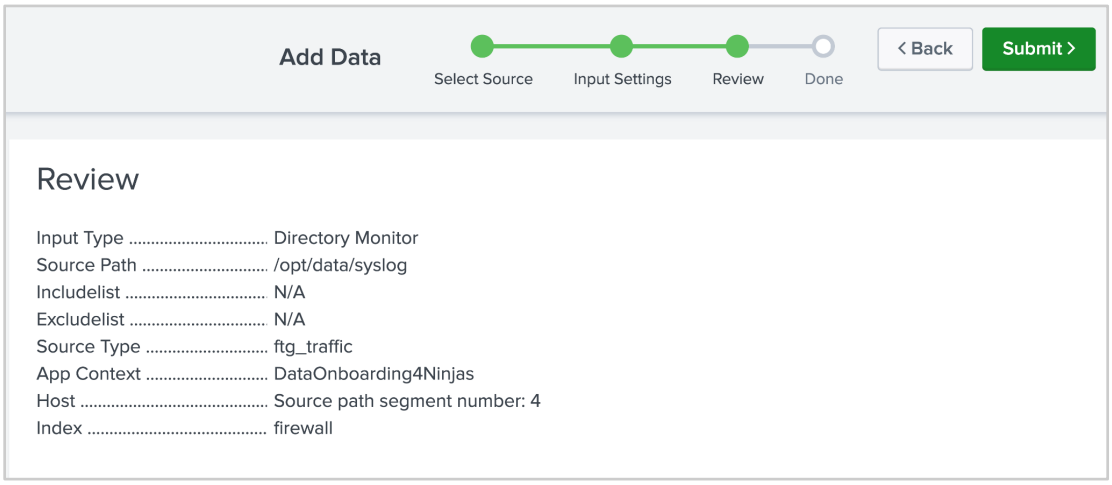


8. Select **Review** in the top right corner.

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Review and Confirm Your Monitoring Input

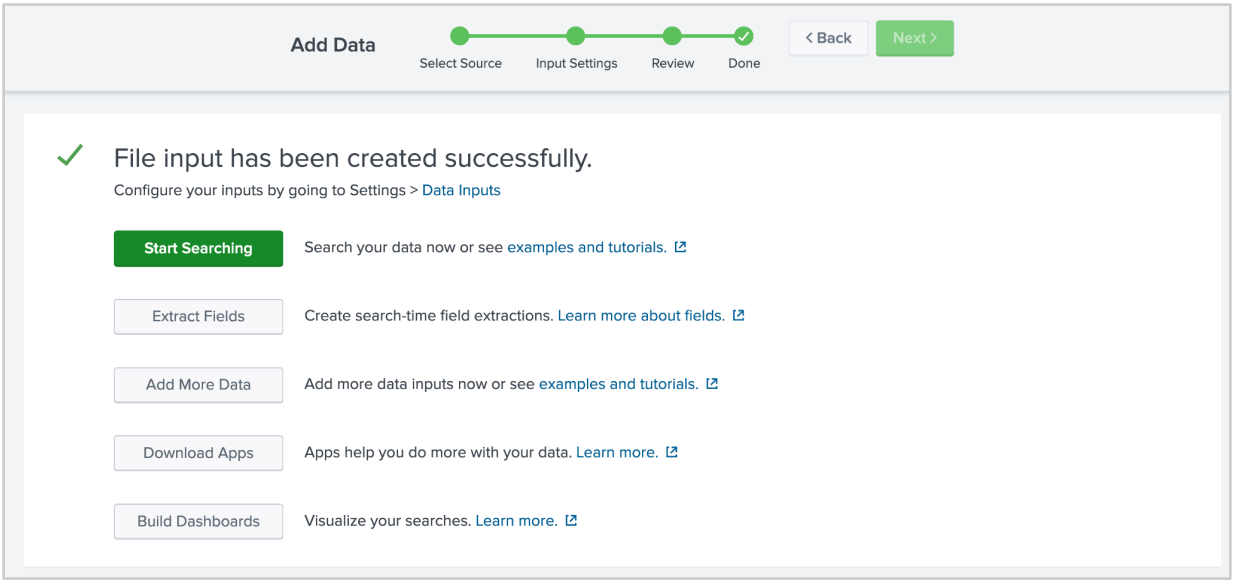
9. Ensure your input information is accurate and click on **Submit**.



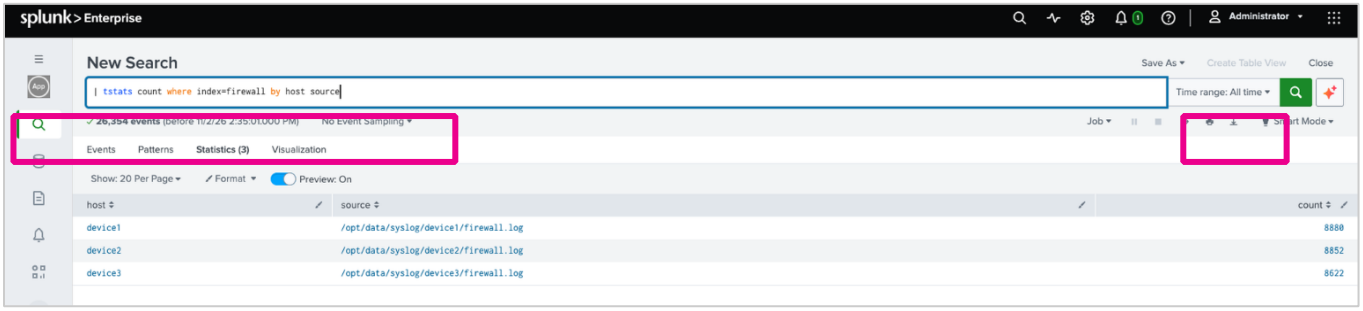
10. Validate your data by clicking on **Start Searching** and entering the following search into the search bar:

```
| tstats count where index=firewall by host source
```

Set the time range picker to **All Time**.



Results should be a table with 3 columns:



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Skip Option 2 if you have already completed Option 1 (GUI).

>> Jump to Lab 2 <<

Option 2: Monitoring Through the CLI

SSH into Your Provided Machine

---

1. Open the Terminal app / CLI
2. Use your given SSH command to log into your individual machine.

🔑 **SSH Details**

The specific IP address for your instance will either be provided to you by the workshop host or will be presented alongside your other workshop instance information within the Splunk Show portal.

If you had to enroll in today's workshop (via Splunk Show) then please log in to https://show.splunk.com to locate your unique SSH command.

Example SSH details:

sshPass	ssh
Sp1unkH00di3	ssh -p 2222 splunk@100.25.43.53

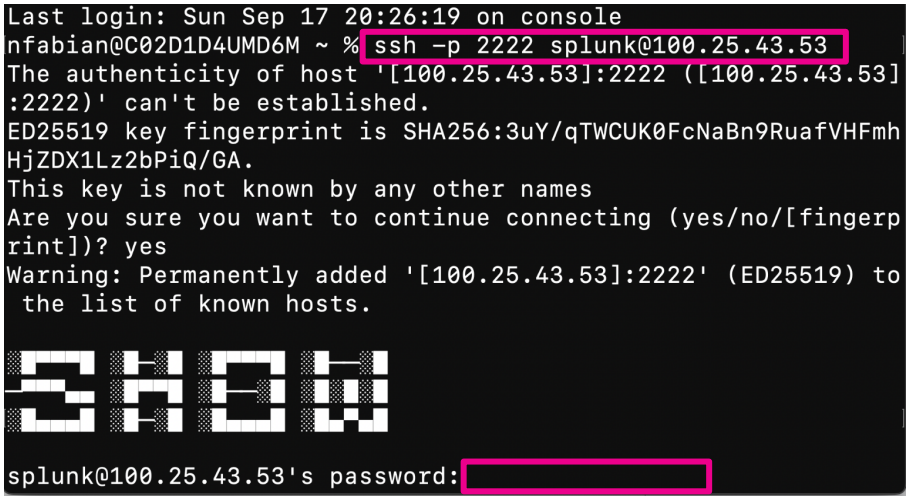
Example SSH command to copy/paste:

```
ssh -p 2222 splunk@100.25.43.53
```

Note: Replace the IP address with the IP address of your workshop instance.

When prompted, enter the SSH password you were provided (this is different from the Splunk login password.)

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3. Upon entering the correct password, you should be logged into the remote machine:



Monitor the Syslog Directory Through inputs.conf

4. Firstly, we will view the current **inputs.conf** settings. Start by switching to the root user:

```
sudo su
```

5. Navigate to the **/opt/splunk/etc/apps/DataOnboarding4Ninjas** directory and list the underlying directories:

```
cd /opt/splunk/etc/apps/DataOnboarding4Ninjas/

ls
```

We should see three directories listed: **bin**, **default** and **metadata**:

```
splunk@show-demo-i-05d1683d07b1bccca:~$ sudo su
root@show-demo-i-05d1683d07b1bccca:/home/splunk# cd /opt/splunk/etc/apps/DataOnboarding4Ninjas/
root@show-demo-i-05d1683d07b1bccca:/opt/splunk/etc/apps/DataOnboarding4Ninjas# ls
bin  default  metadata
```

6. Splunk app configuration changes should ONLY be made in the **local** directory of an app. Since the local directory does not yet exist, we need to create one:

```
mkdir local
```

Note: Ensure the folder name is all lowercase

```
root@show-demo-i-05d1683d07b1bccca:/opt/splunk/etc/apps/DataOnboarding4Ninjas# mkdir local
root@show-demo-i-05d1683d07b1bccca:/opt/splunk/etc/apps/DataOnboarding4Ninjas# ls
bin  default  local  metadata
```

7. We will now add a monitoring stanza to the **inputs.conf** file to instruct Splunk to monitor our firewall log files. To do this, navigate to the newly created **local** directory and open the **inputs.conf** file to edit:

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```
cd local

vi inputs.conf
```

```
root@show-demo-i-05d1683d07b1bccca:/opt/splunk/etc/apps/DataOnboarding4Ninjas# cd local
root@show-demo-i-05d1683d07b1bccca:/opt/splunk/etc/apps/DataOnboarding4Ninjas/local# vi inputs.conf
```

📝 **vi (Visual Editor)**

vi is a text editor used particularly for UNIX machines. There are many cheat sheets available online to help you navigate the commands when working with the editor. There are also other options for editing Splunk .conf files in Linux, such as nano.

---

8. Press the " **i** " key to start editing the file:

```
i
```

9. Within the configuration file, type or paste in the following stanza:

```
[monitor:///opt/data/syslog/*/firewall.log]

sourcetype = fgt_traffic

index = firewall

host_segment = 4
```

ℹ️ This stanza defines the input as firewall logs within the syslog directory. The data is being assigned a sourcetype of **fgt_traffic** and sent to the **firewall** index and assigning the host name to the **4th segment of the directory path**. In this case the host hame will be assigned whatever directory follows **/syslog/** in the stated path.

---

10. Exit the vi editor by pressing the **escape** key then typing **:wq** and hitting **enter**. This is telling the editor to save and exit back to the main CLI.

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Reload and confirm the monitoring input

11. Reload the monitor input and restart Splunk:

```
/opt/splunk/bin/splunk reload monitor

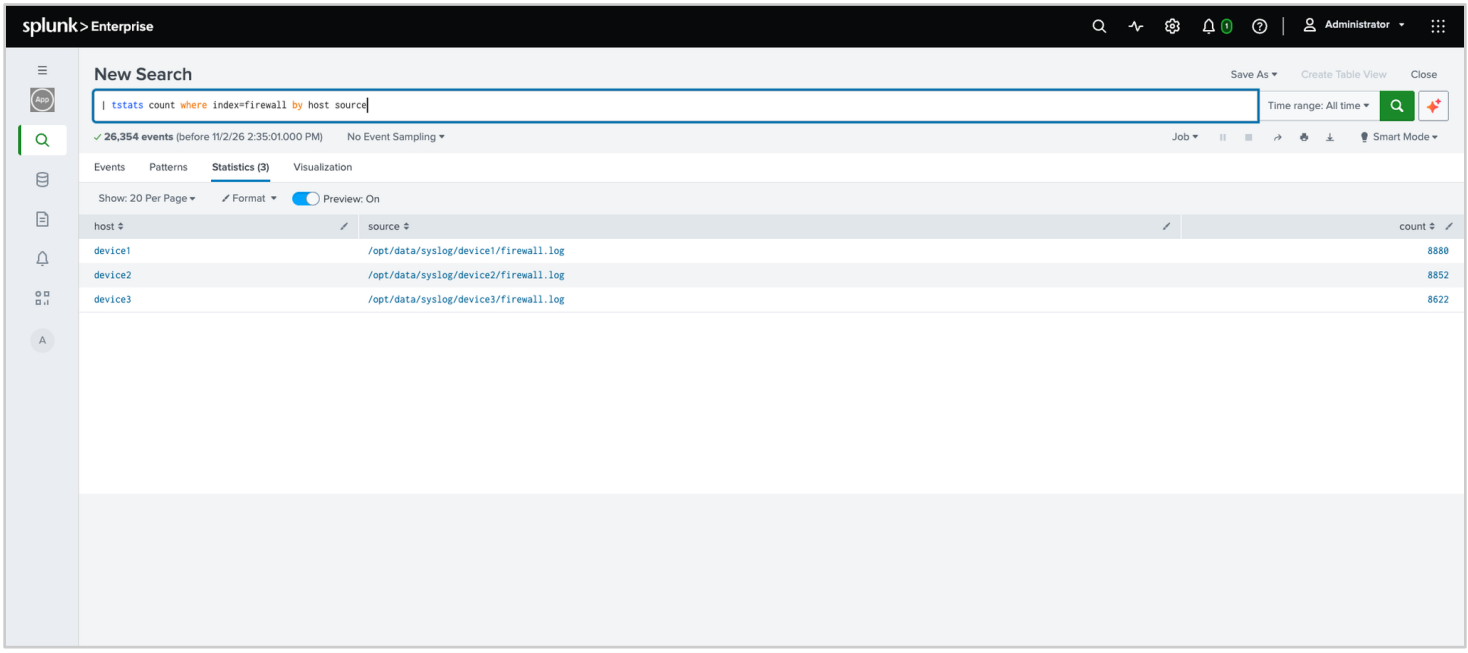
sudo /opt/splunk/bin/splunk restart
```

```
root@show-demo-i-05d1683d07b1bccca:/opt/splunk/etc/apps/DataOnboarding4Ninjas/local#
 /opt/splunk/bin/splunk reload monitor
WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[
sslConfig]/cliVerifyServerName for details.
Your session is invalid.  Please login.
Splunk username: admin
Password:
Monitor inputs reloaded
root@show-demo-i-05d1683d07b1bccca:/opt/splunk/etc/apps/DataOnboarding4Ninjas/local#
/opt/splunk/bin/splunk restart
Stopping splunkd...
```

12. We now need to confirm that events are being ingested. To do this, log back in to Splunk (you will have been logged out due to the restart command we ran above) and navigate to **Apps > DataOnboarding4Ninjas**. Run the following search over **All Time**:

```
| tstats count where index=firewall by host source
```

13. Results should be a table with 3 columns:



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Lab 2 - Data Parsing

Description

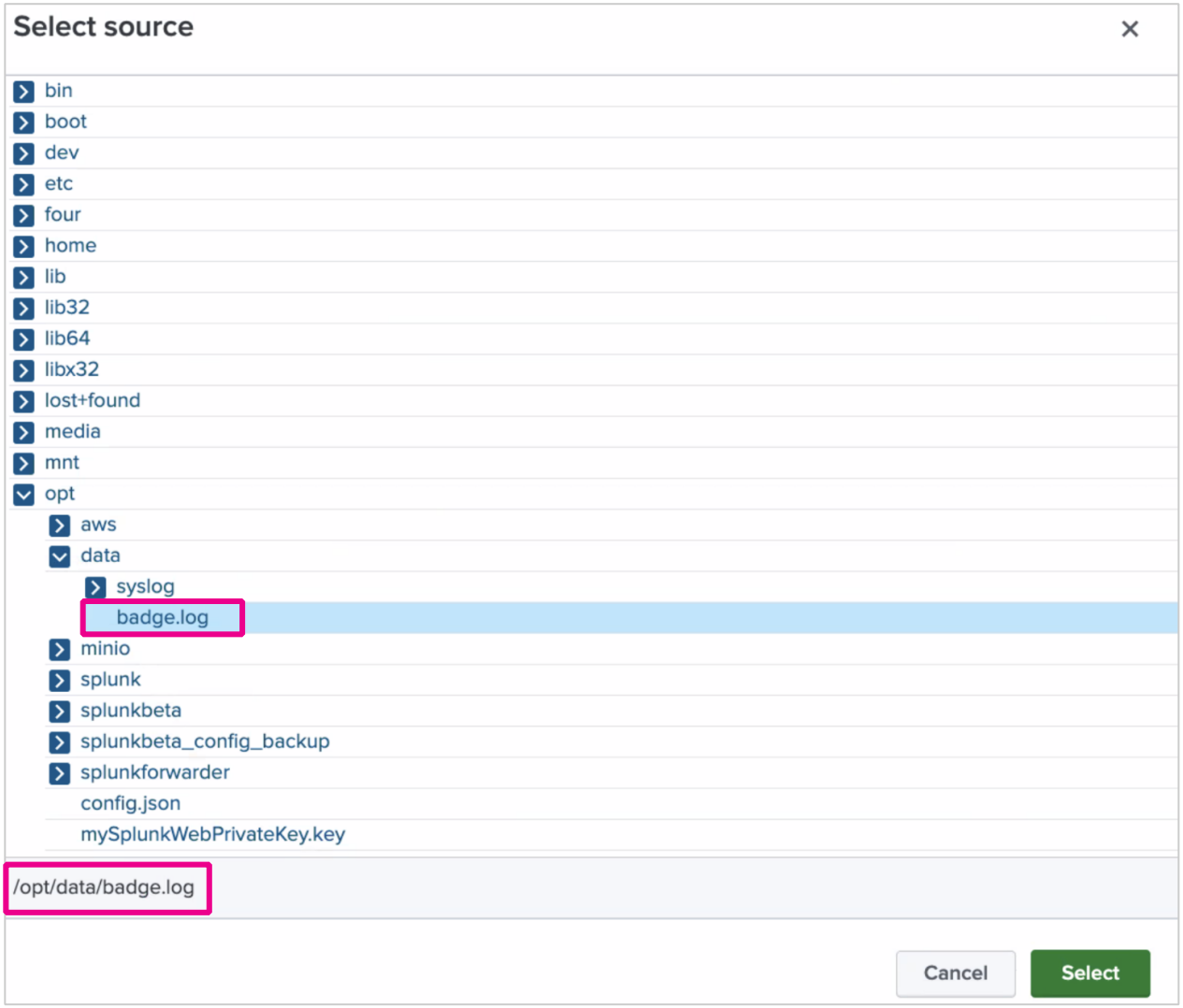
In this activity, we ingest badge.log data into Splunk a single time (rather than monitoring continuously). We will set up a source type to correctly parse the data into individual events and then we will pivot into the Splunk Add-on Builder app where we import our source type into a new Add-on named 'badge'.

This lab will be completed in the UI but several tasks can also be done in the CLI. See the Appendix Optional Activity on how tasks 1-4 can be accomplished through the CLI.

Select the badge.log File as your Source

---

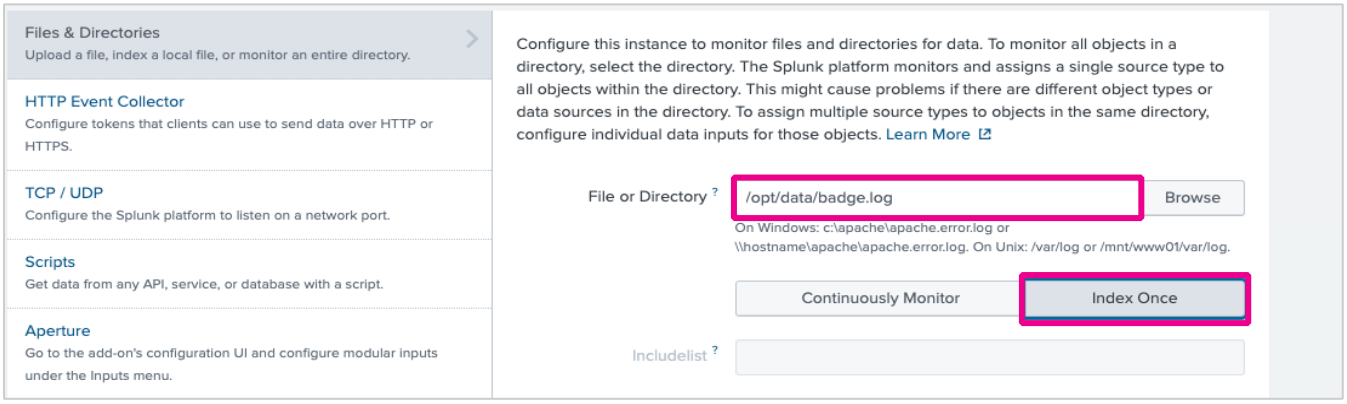
1. Navigate to **Settings > Add Data > Monitor > Files & Directories** and click on **Browse**.
2. Browse to **opt > data** and select **badge.log**.



3. Ensure your **File or Directory** field now contains **/opt/data/badge.log**.

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4. Select **Index Once** and click on **Next**.



ℹ️ Index Once

Selecting **Index Once** is equivalent to a 'oneshot' command in the CLI. This tells Splunk to read the file once, rather than establishing a continuous monitor of the file. This could be useful if there are segregated systems offline that cannot connect to Splunk; you can download the file and ingest it one time to enrich data or help investigate. This is also useful when testing data!

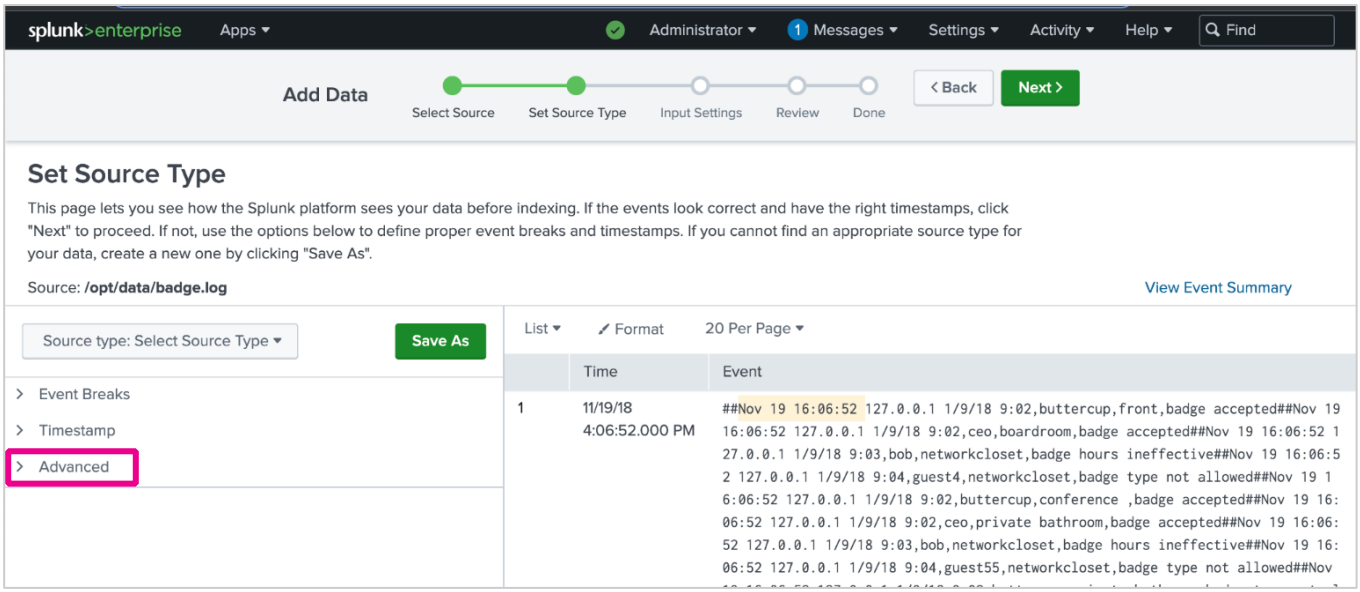
Set the Source Type

5. We now need to verify our data parsing settings. In the data preview pane on the right of the **Set Source Type** page we can see that the data is not getting ingested in the correct format; it is coming in as one big event rather than individual events, so we need to set a new source type to rectify this.

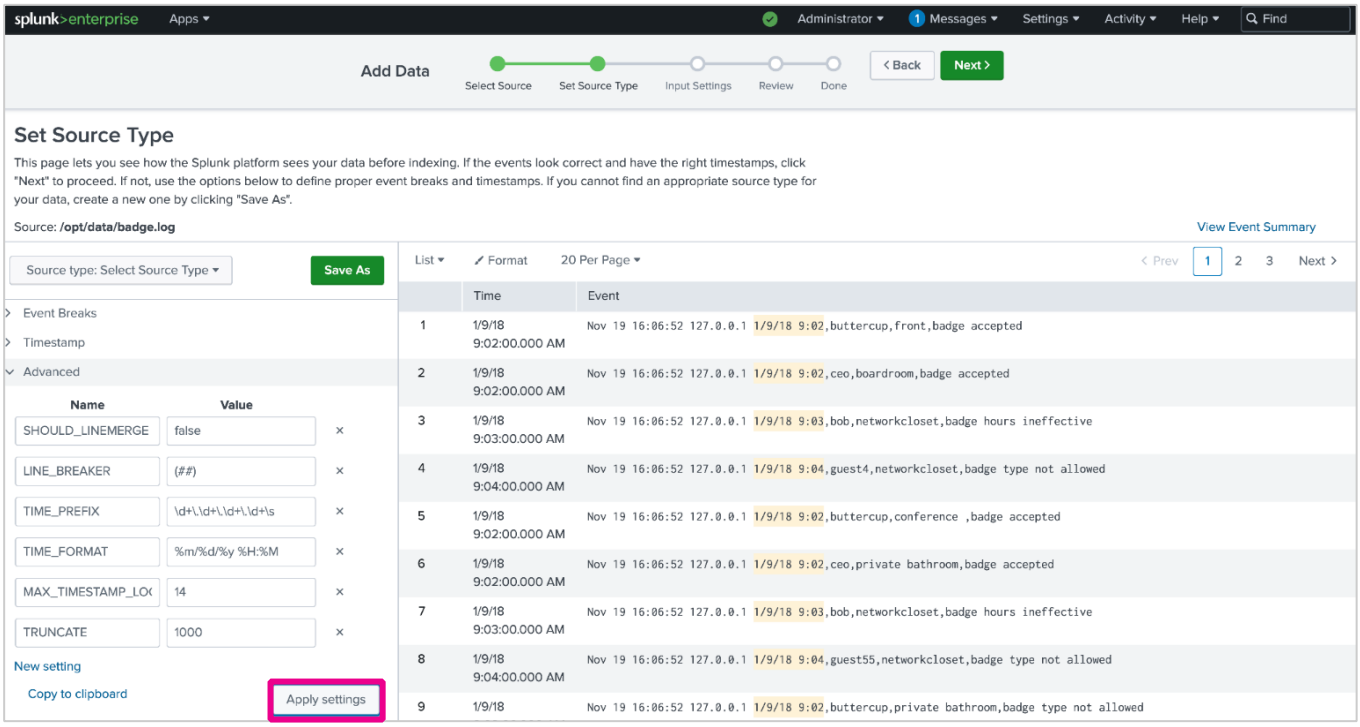
   On the left of the page, expand out the **Advanced** section and adjust the source type settings to match the following:

Name	Value
SHOULD_LINEMERGE	false
LINE_BREAKER	(##)
TIME_PREFIX	\d+\.\d+\.\d+\.\d+\s
TIME_FORMAT	%m/%d/%y %H:%M
MAX_TIMESTAMP_LOOKAHEAD	14
TRUNCATE	1000

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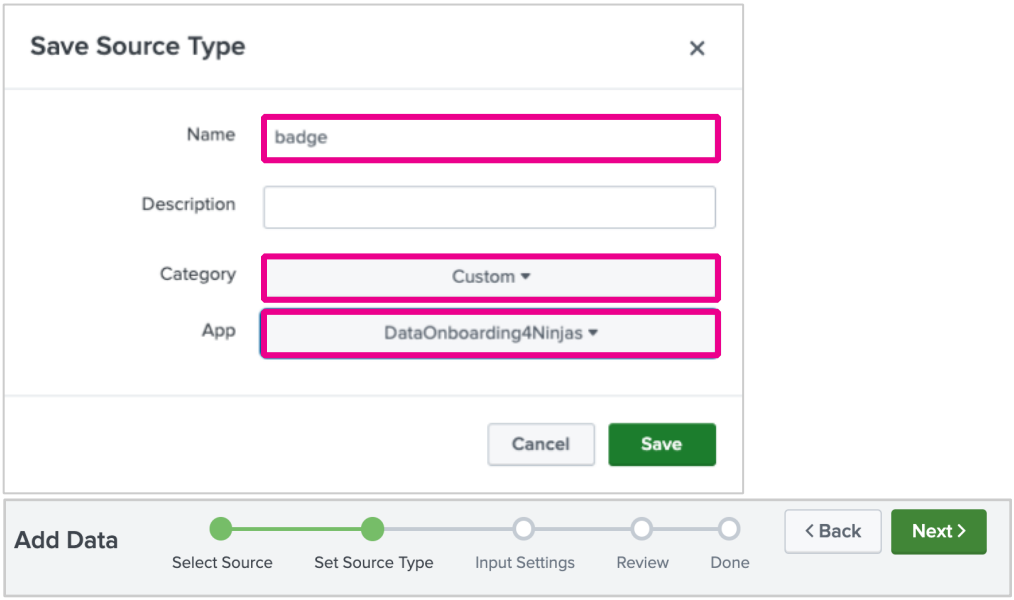
6. Click on **Apply Settings** - the single event will now be broken into individual events.



7. Now save the new source type by clicking on **Save As**.
8. For the source type name enter "**badge**".
9. Set the **Category** as **Custom**.
10. For the **App** select **DataOnboarding4Ninjas**.

    Click **Save** and **Next**.

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Saving Source Types

It is best practice to save source types to the specific apps they operate within rather than having them saved to the default Search & Reporting app, which is why we select DataOnboarding4Ninjas.

After saving, there may be additional settings added to the source type. For the purposes of this lab, we do not need to worry about these.

11. Upon saving, a badge sourcetype will now be visible in the props.conf for the DataOnboarding4Ninjas app. The props.conf file can be found in the directory **/opt/splunk/etc/apps/DataOnboarding4Ninjas/local**.

To view the contents and see the new **[badge]** stanza that has been added, run the following CLI commands:

```
cd /opt/splunk/etc/apps/DataOnboarding4Ninjas/local

cat props.conf
```

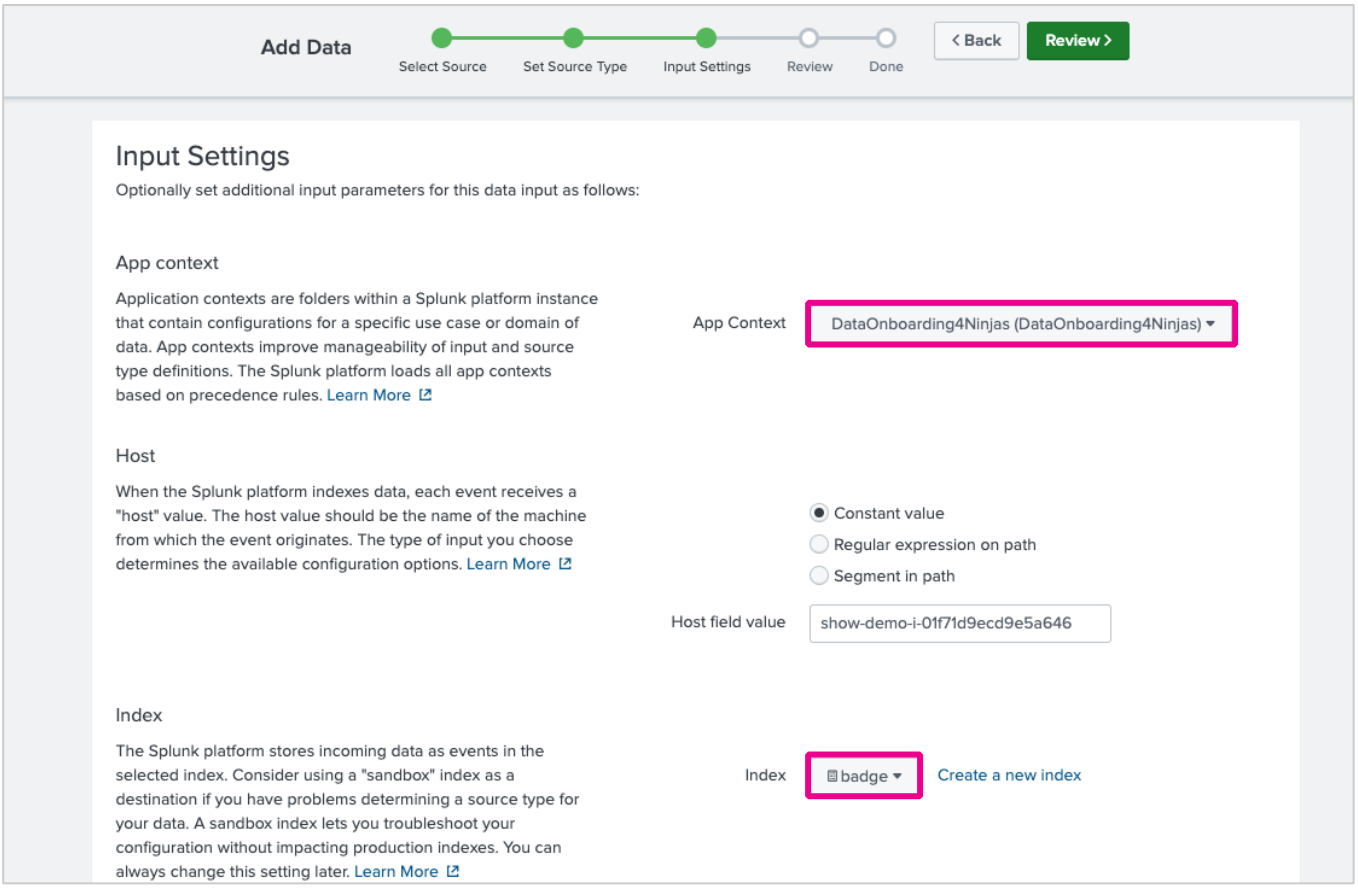
Adjust the Input Settings

12. On the **Input Settings** screen, set the **App Context** to **DataOnboarding4Ninjas.**
13. Set the **Index** to **badge**.

The badge Index

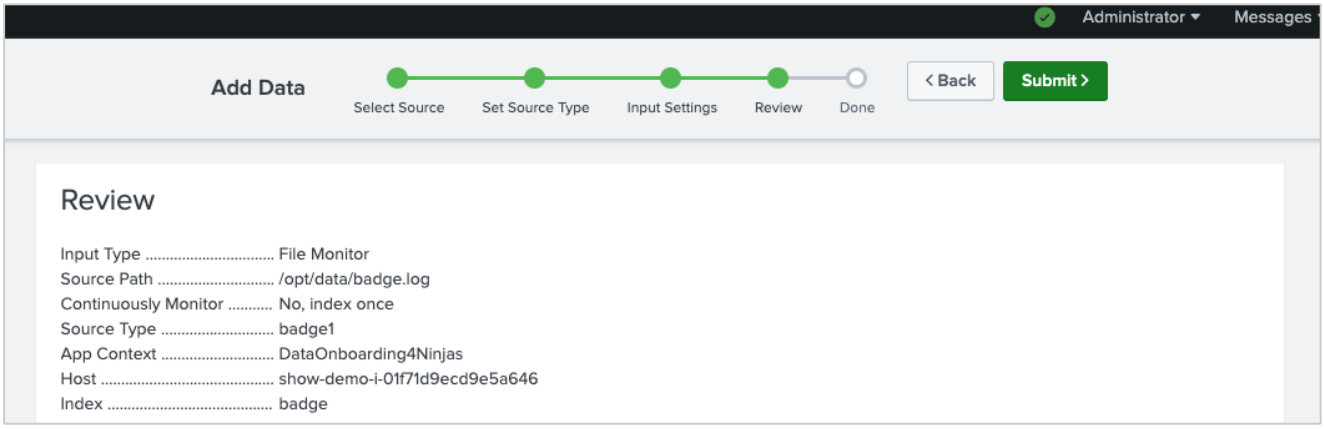
In this environment, we already have the badge index created. In production, ensure that you first create the proper index so you can route your data there when setting up data ingestion. Please note that it is not best practice to send your data to the default “main” index.

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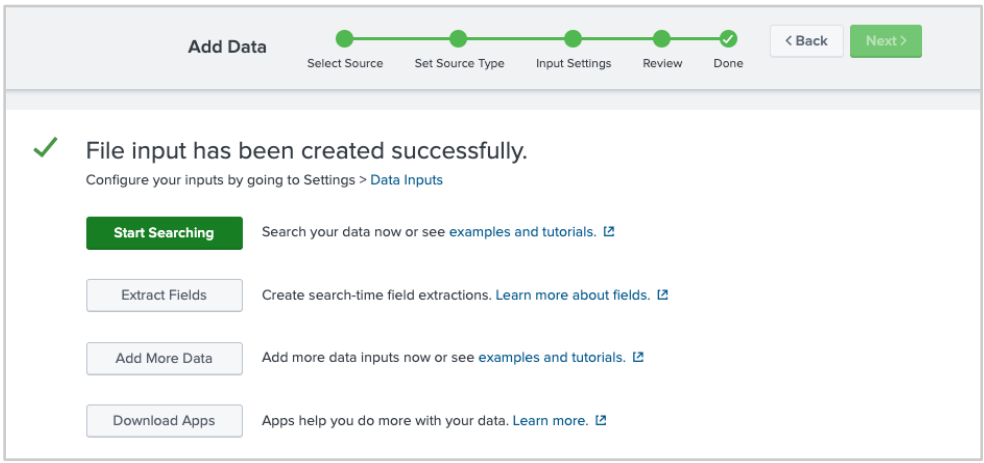
Submit and Search Your Data

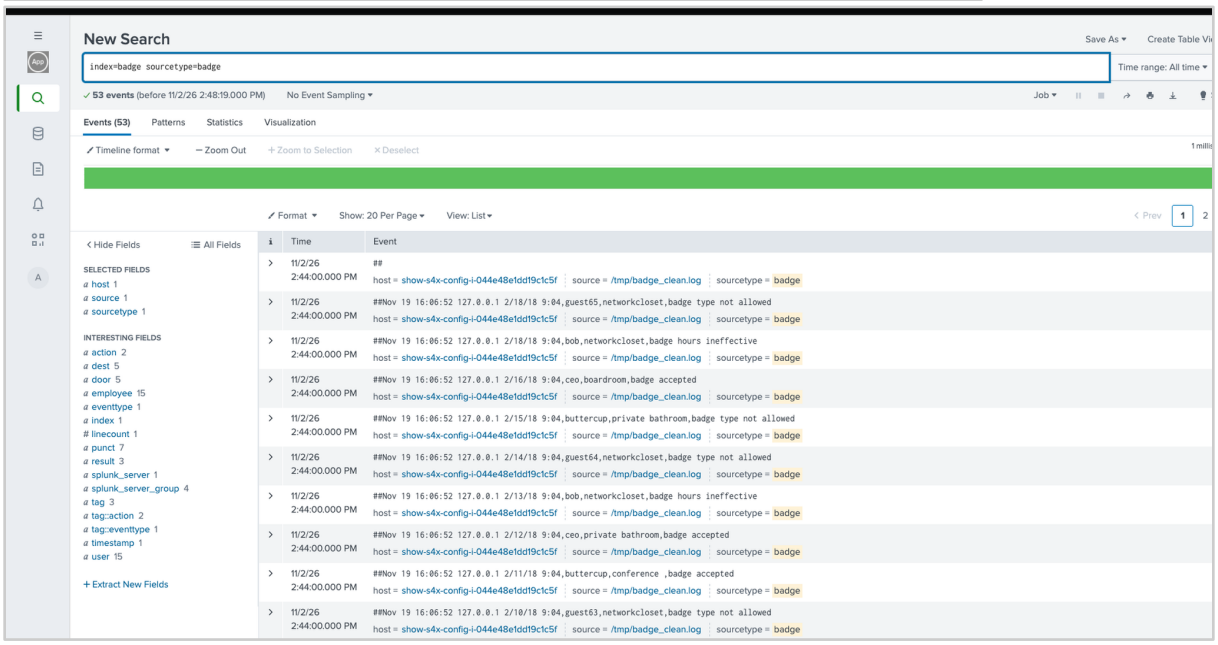
14. With the input settings adjusted, you can select **Review** and then **Submit** on the next page.



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15. On the **File input has been created successfully** page click on **Start Searching**. This will take you to a new search - with the search query populated - where you can see your badge.log data now coming in.





Set Up the Add-on Builder

Earlier in this lab we created a source type during the “Add Data” process. Now we are going to import that source type into the Splunk Add-on Builder where we can save it for future use as a Technology Add-on, or “TA”. This way anyone can easily grab that add-on and install it to bring in badge data.

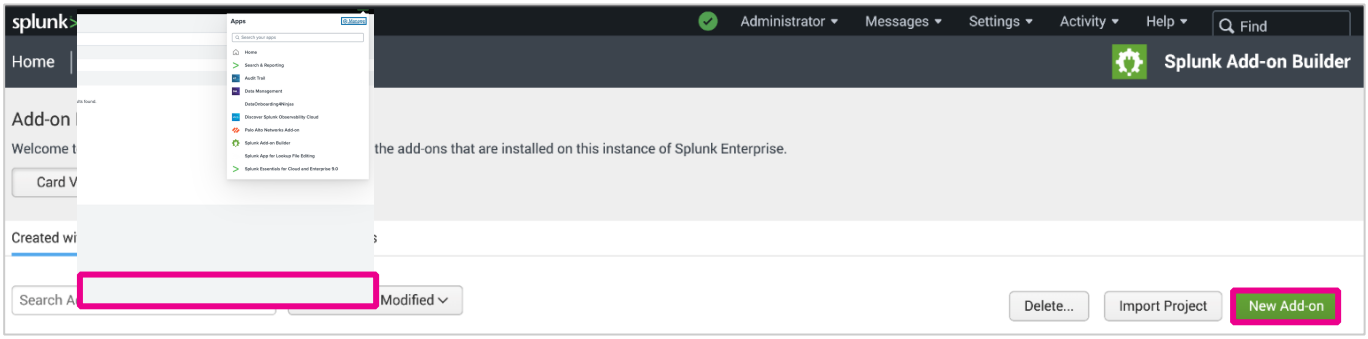
Splunk Add-on Builder

For more information about the Splunk Add-on Builder and Splunk add-ons in general, please see the following helpful resources:

- https://docs.splunk.com/Documentation/AddonBuilder/latest/UserGuide/Overview
- https://docs.splunk.com/Documentation/AddOns/released/Overview/AboutSplunkadd-ons

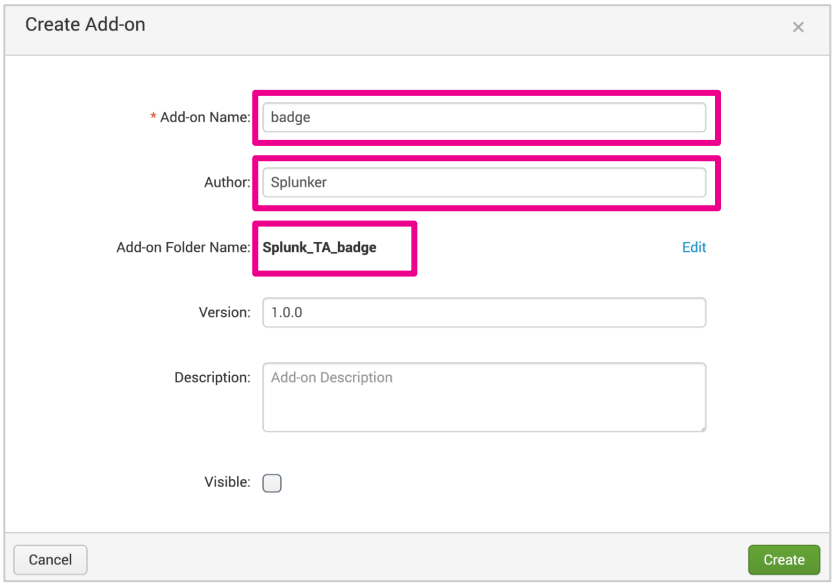
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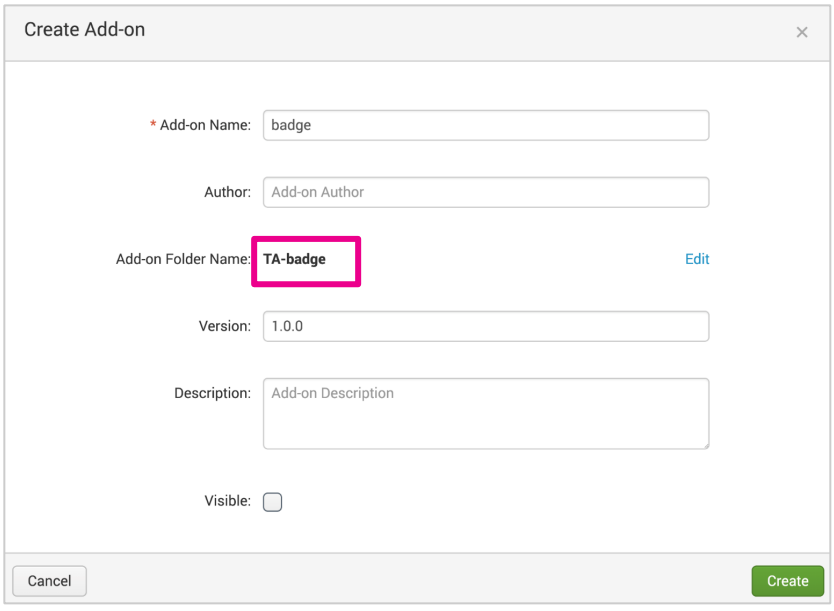
16. Navigate to **Apps > Splunk Add-on Builder** and click on **New Add-on**.



17. For the **Add-on Name** enter "**badge**". For the author you can enter your own name or leave it blank.

    Note the **Add-on Folder Name** that is generated, as this can change based on who the author is. This is the name of the app folder where you will find the underlying configuration files through the CLI.

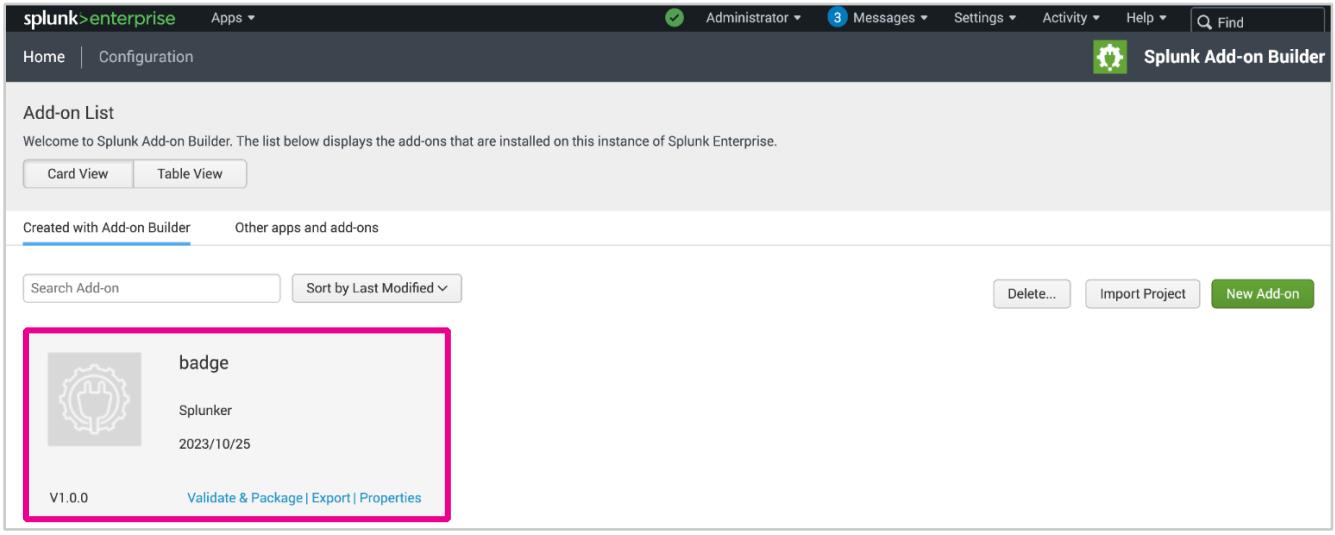




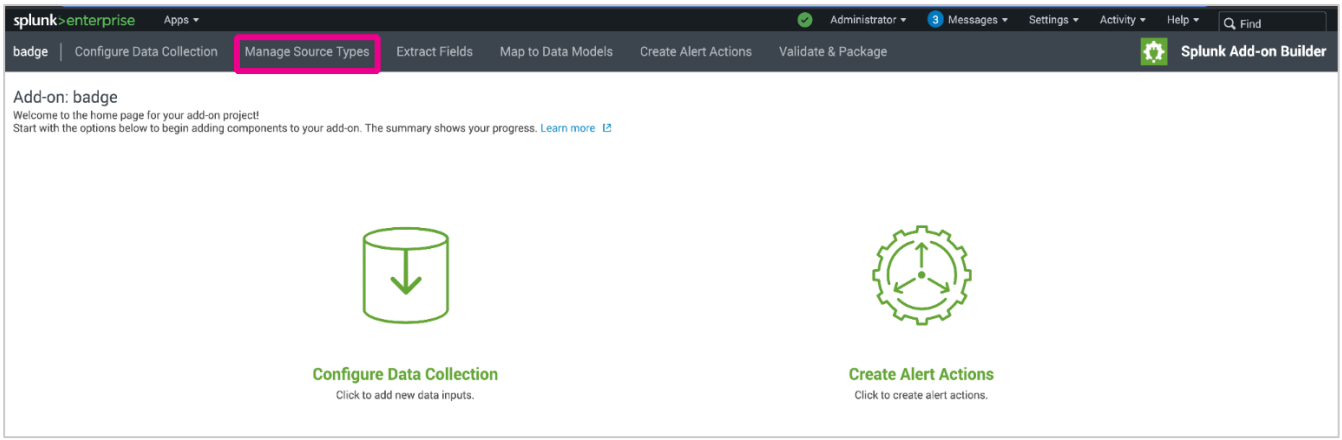
18. Click on **Create**.

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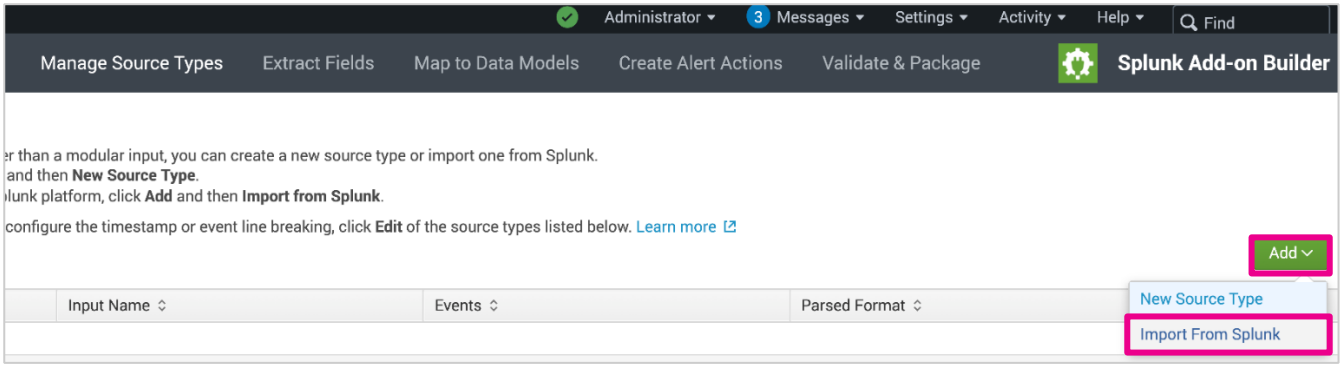
19. You can now see your badge add-on on the home screen of the Splunk Add-on Builder app. Open the badge add-on by clicking on it.



20. We will now import the source type we created earlier so it is contained within the badge add-on. To do this, click on **Manage Source Types** in the menu bar.

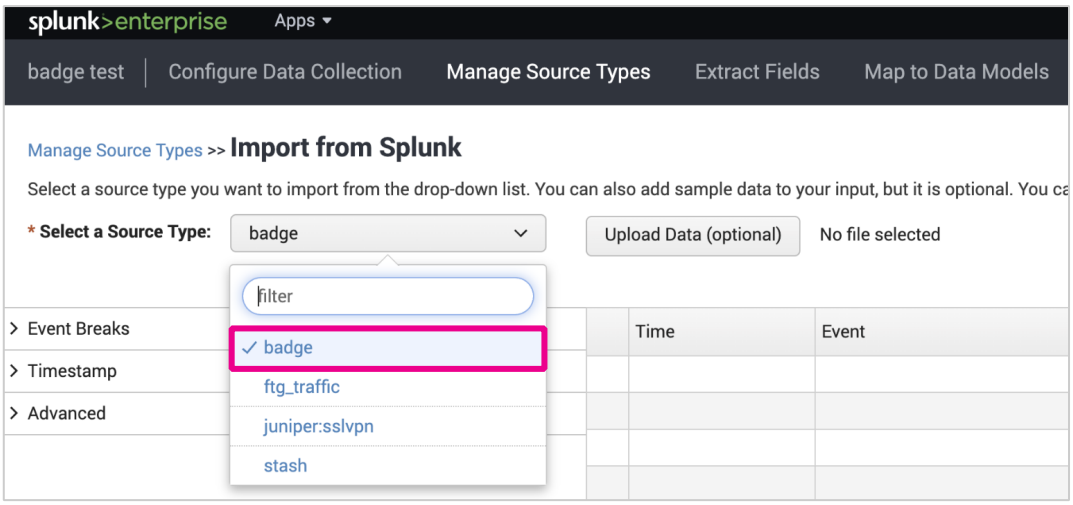


21. Click on the **Add** button and select **Import From Splunk** from the dropdown menu.



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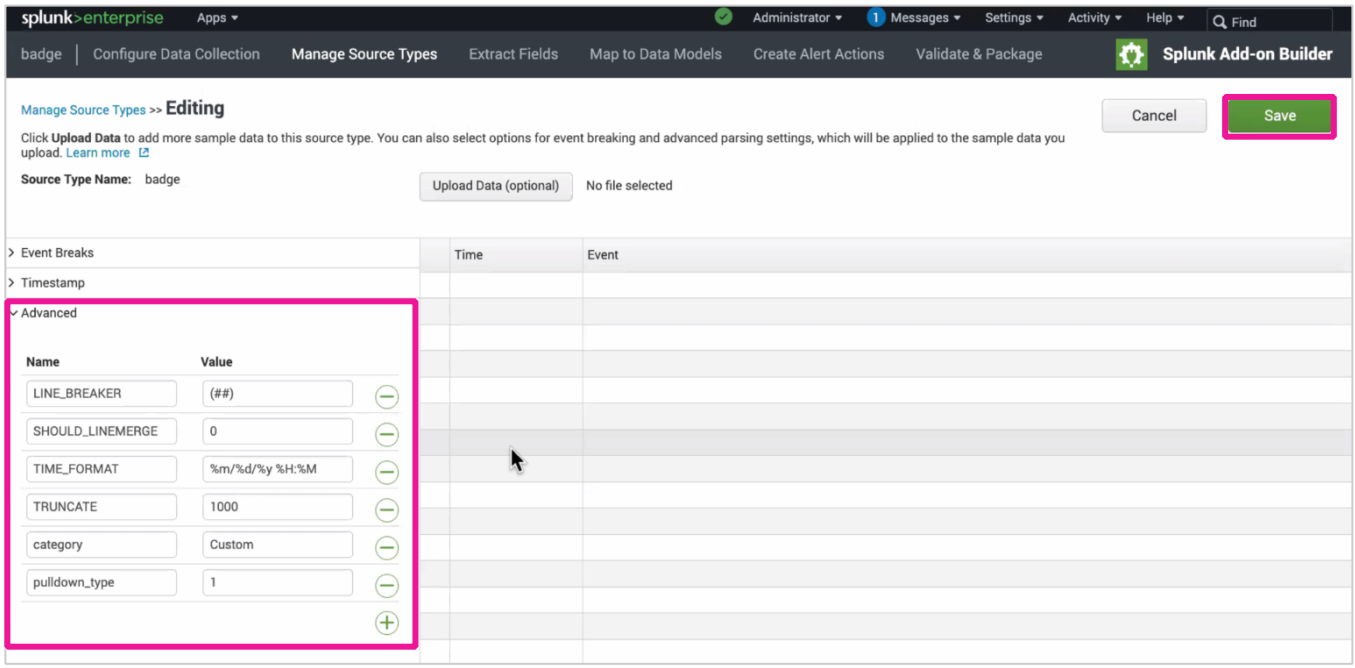
22. Select **badge** from the **Select a Source Type** dropdown.



The sourcetype parameters defined earlier (see Set the Source Type) will now be imported into the add-on builder and will be visible on the left under the **Advanced** section.

The Splunk Add-on Builder imports source type settings but it may also pull in other default settings. You can always continue to adjust your settings here as well.

Expand out the **Advanced** section on the left of the page and confirm that the field name and values we defined earlier are present.



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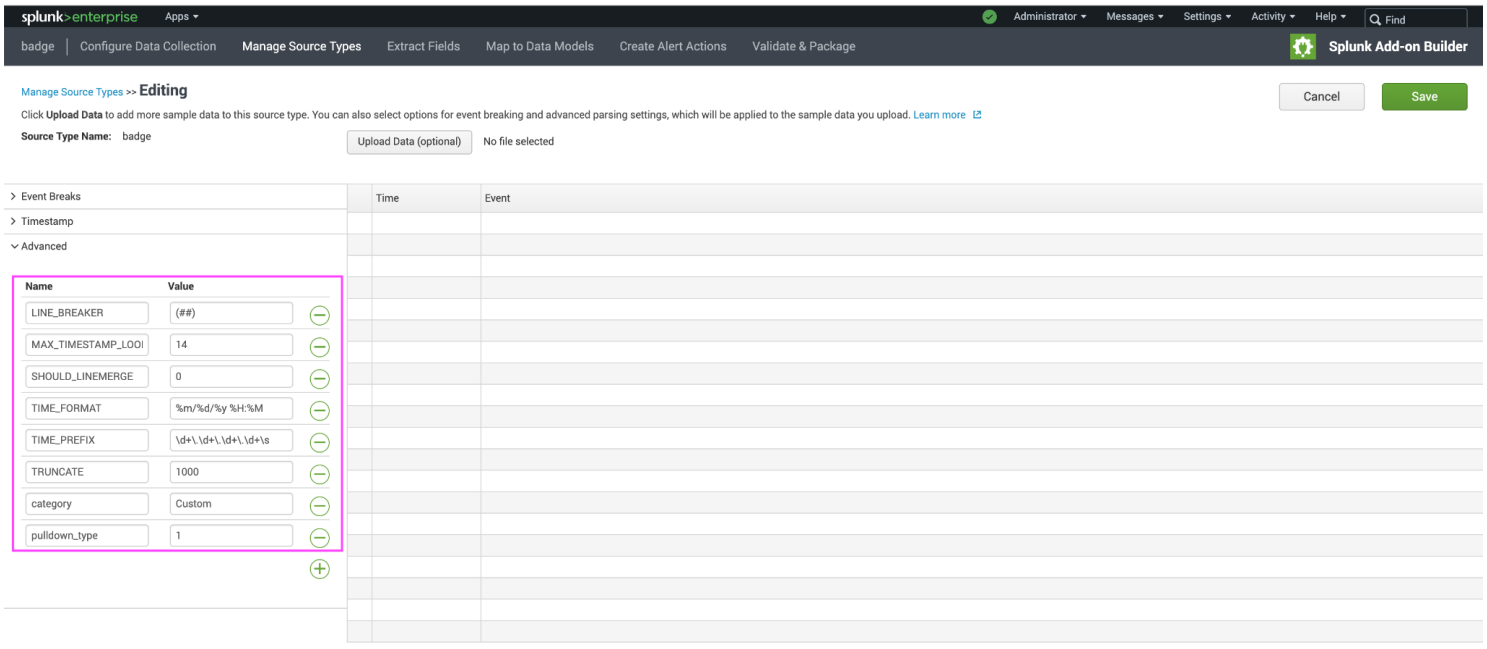
🔑 You may have missing fields

As you can see in the screenshot above there are at least 2 fields from our original sourcetype definition missing. This may or may not be true when you are using this lab but please check as it can affect end results.

Again, here is that list:

Name	Value
SHOULD_LINEMERGE	false
LINE_BREAKER	(##)
TIME_PREFIX	\d+\.\d+\.\d+\.\d+\s
TIME_FORMAT	%m/%d/%y %H:%M
MAX_TIMESTAMP_LOOKAHEAD	14
TRUNCATE	1000

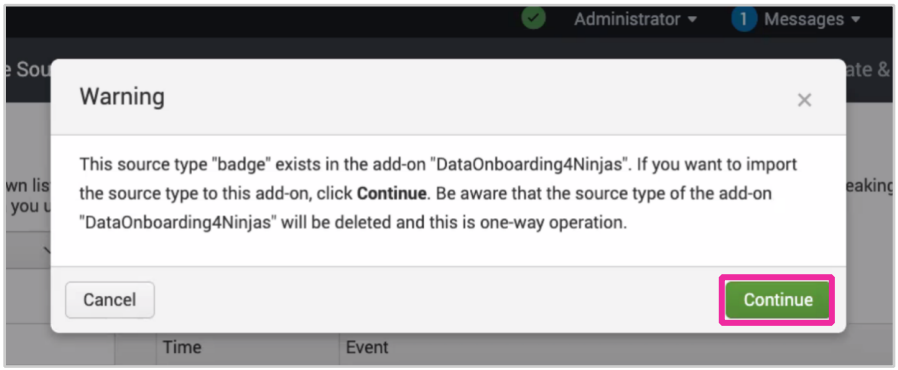
Please be sure to add them and ensure your values look something list this:



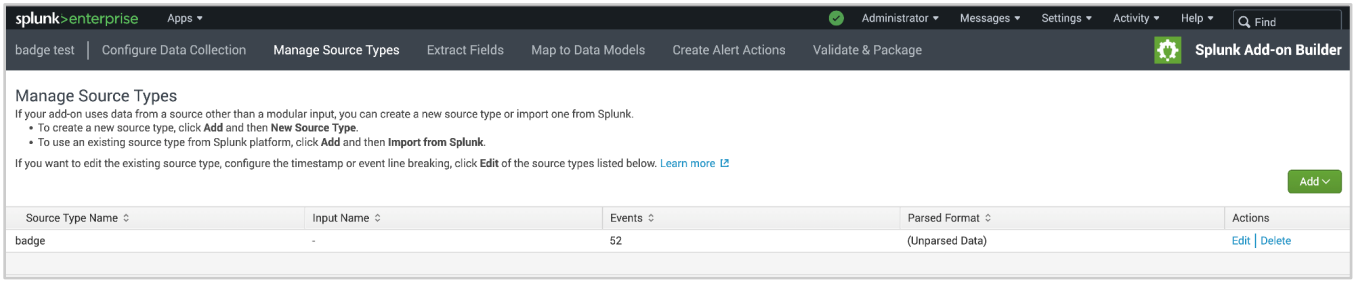
23. Click **Save**.

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24. On the popup **Warning** message, click on **Continue**. The Warning is letting you know that the source type will no longer be associated with the **DataOnboarding4Ninjas** app but will instead be associated with the Splunk Add-on Builder app. Since this is the intention we can ignore the warning.



25. Click **Save** to navigate back to the **Manage Source Types** home screen.



ℹ️ Add-on Folders

As previously mentioned, when creating a new add-on within the Splunk Add-on Builder (see step 18), an **Add-on Folder Name** directory will be created under the Splunk **/etc/apps** directory. Then, when importing and saving a sourcetype within the new add-on (step 24) Splunk will save that sourcetype within a local props.conf of the TA.

You can now find your badge add-on listed as **/opt/splunk/etc/apps/<TA FOLDER NAME>** and your updated **[badge]** sourcetype within props.conf under **/opt/splunk/etc/apps/<TA FOLDER NAME>/local/props.conf**

Command to see a list of Splunk apps where your badge add-on sits:

```
ls /opt/splunk/etc/apps
```

Command to see your new props.conf file:

```
sudo cat /opt/splunk/etc/apps/<TA FOLDER NAME>/local/props.conf
```

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Lab 3 - Field Extraction and CIM Compliance

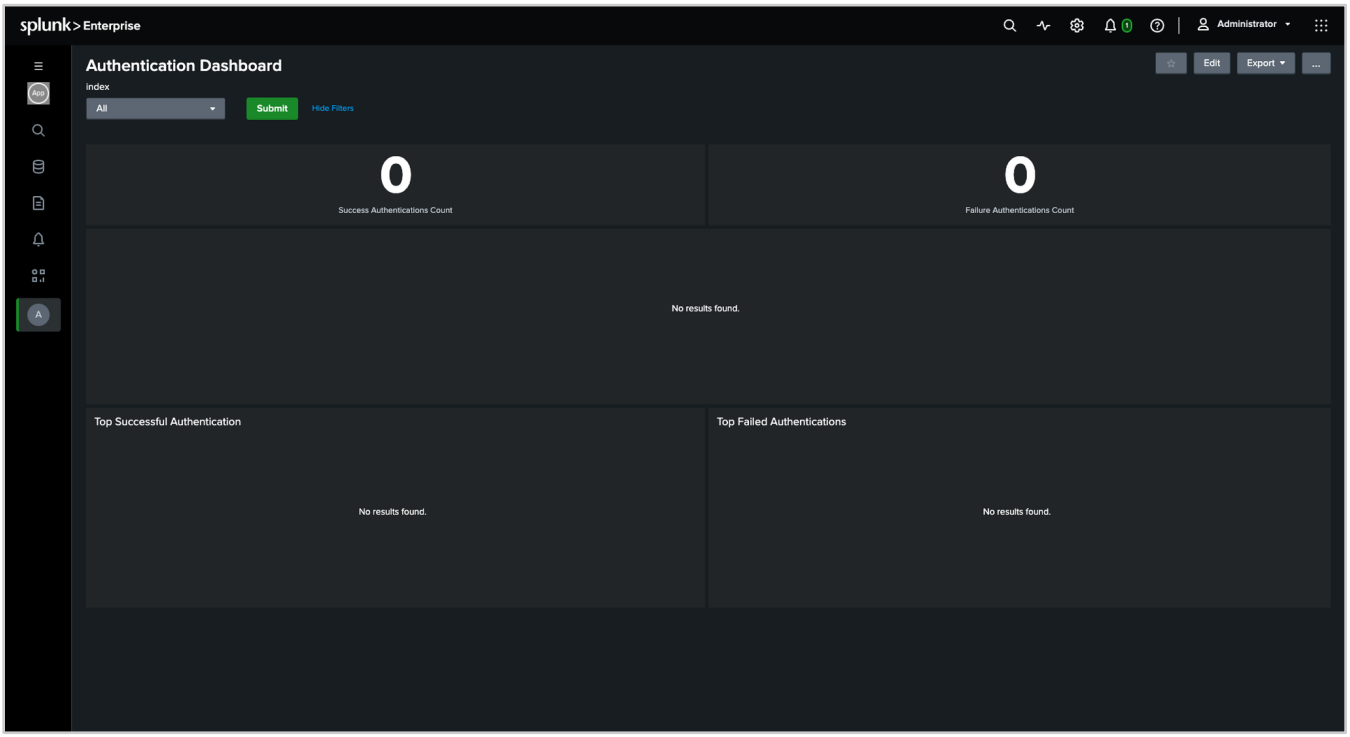
Description

In this activity we configure our badge add-on to extract fields from our badge.log data and we map our data to the Authentication data model available through Splunk's Common Information Model (CIM). In mapping our data to the data model, the Authentication Dashboard available in the DataOnboarding4Ninjas app will start to populate.

Validate the Empty Authentication Dashboard

1. Navigate to **Apps > DataOnboarding4Ninjas** and click on **Authentication Dashboard** in the menu bar.

There should be no content populating the dashboard yet.



Although we have badge authentication data ingested, the searches written for this dashboard are looking for information tied to the Authentication data model. This data model is available as part of the Splunk Common Information Model (CIM). We are going to map our badge TA to the authentication data model so the searches in the dashboard know to pick up the badge data we have coming in.

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ℹ️ Data Models

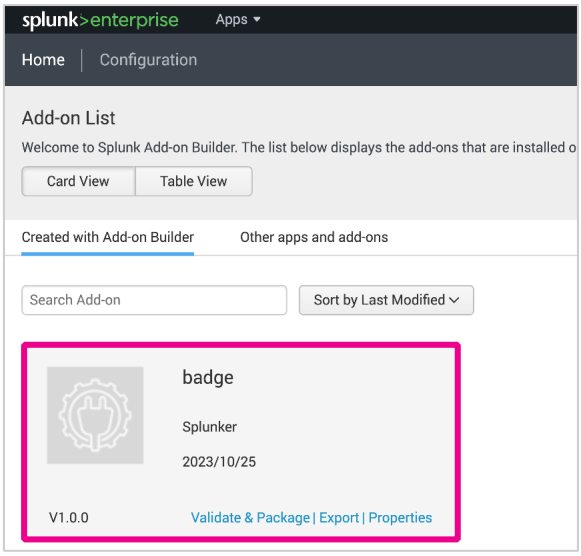
Further information about data models can be found in the Knowledge Manager Manual:
https://docs.splunk.com/Documentation/Splunk/latest/Knowledge/WhatisSplunkknowledge

To learn about the Splunk Common Information Model (CIM) please see:
https://docs.splunk.com/Documentation/CIM/latest/User/Overview

Further information about the Authentication data model specifically can be found here:
https://docs.splunk.com/Documentation/CIM/latest/User/Authentication

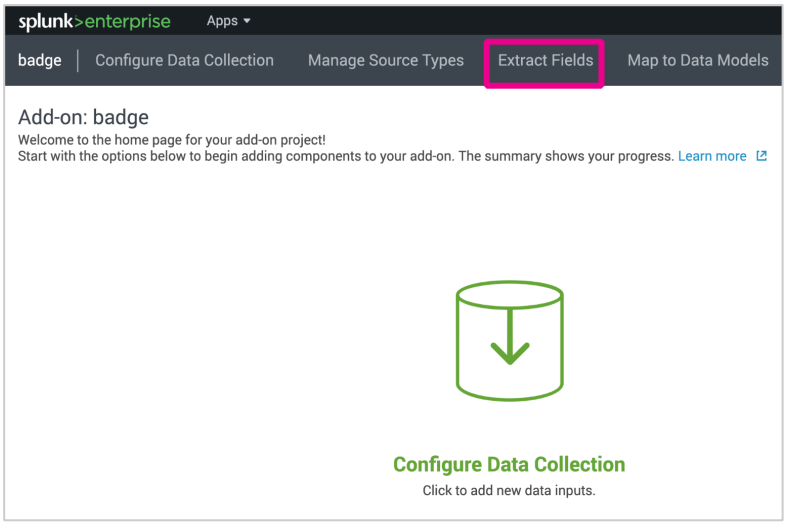
Extract Fields from the Badge Data

2. Navigate to the **Splunk Add-on Builder App**. On the **Add-on List** page click on the **badge** add-on.



3. In the top menu click on **Extract Fields**.

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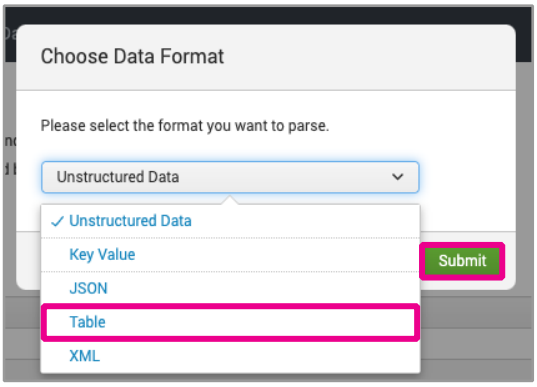


4. We will now establish a table format to help parse the data.

To do this, on the **Extract Fields** page you will see your badge source type listed. Under **Actions**, click **Assisted Extraction**.

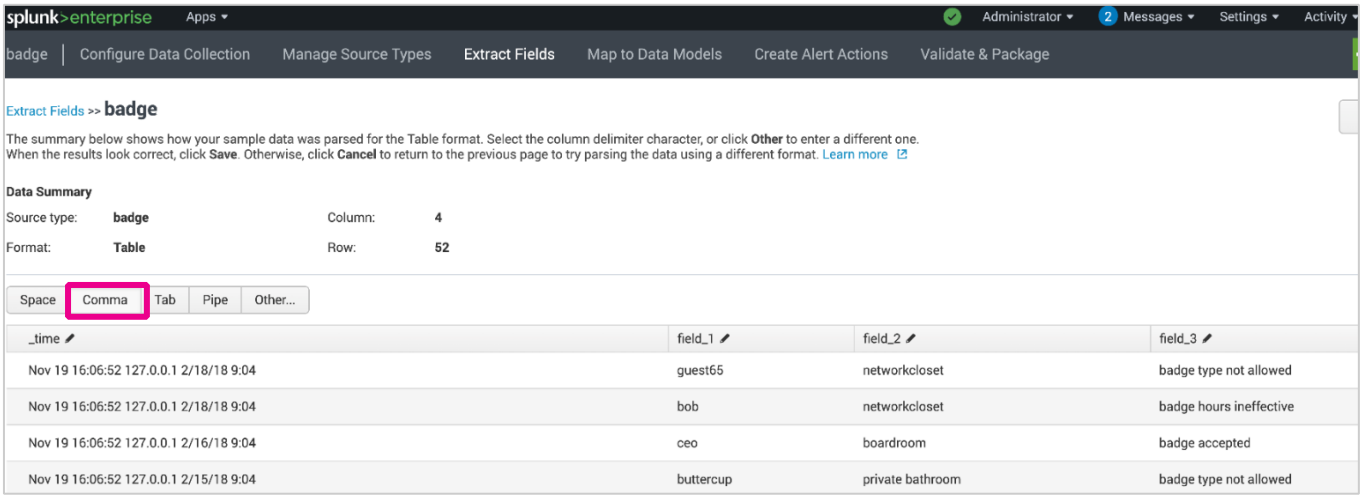


5. On the **Choose Data Format** popup, select the **Table** data format and click **Submit**.



6. Results are returned broken down by delimited fields in a table format. Select **Comma** as the separator.

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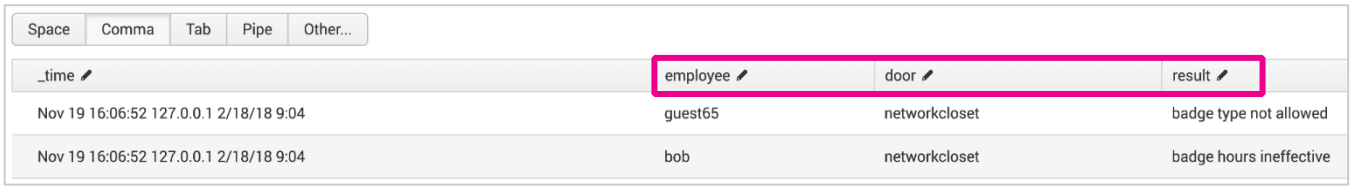


7. Add a field name as the header of each column:

field_1 = **employee**

field_2 = **door**

field_3 = **result**



Optional Step - view the underlying .conf files that get built as we adjust the Add-on in the GUI

8. Prior to saving, view your local directory configuration files via the CLI. List the files in the local directory of your TA by running the following command:

```
ls /opt/splunk/etc/apps/<TA_FOLDER_NAME>/local
```

There is an **app.conf** and a **props.conf** file listed, but currently no **transforms.conf**. Upon saving the field names as the extraction, a transforms.conf will be created to store the configurations.

```
splunk@show-demo-i-0a2741c4943a40369:~$ ls /opt/splunk/etc/apps/TA-badge/local
app.conf  props.conf
splunk@show-demo-i-0a2741c4943a40369:~$
```

9. Back in the GUI, click **Save** in the Splunk Add-on Builder.

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Optional Step - view the underlying .conf files that get built as we adjust the Add-on in the GUI

10. Verify the transforms.conf file by running the following CLI command to list the files in the **local** directory of your TA:

```
ls /opt/splunk/etc/apps/<TA_FOLDER_NAME>/local
```

```
splunk@show-demo-i-0a2741c4943a40369:~$ ls /opt/splunk/etc/apps/TA-badge/local
app.conf  props.conf  transforms.conf
splunk@show-demo-i-0a2741c4943a40369:~$
```

11. View the **transforms.conf** and see the new field names defined:

```
sudo cat /opt/splunk/etc/apps/<TA_FOLDER_NAME>/local/transforms.conf
```

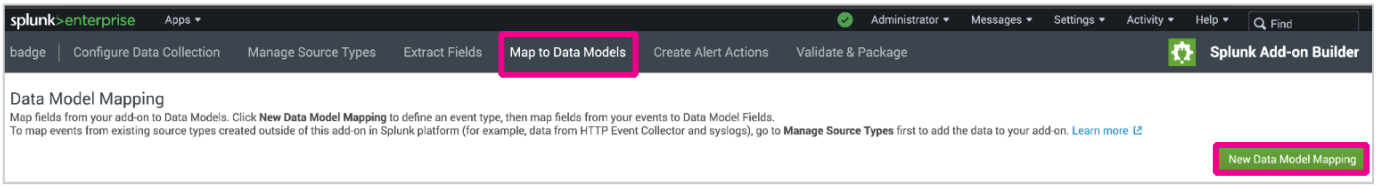
```
splunk@show-demo-i-0a2741c4943a40369:~$ sudo cat /opt/splunk/etc/apps/TA-badge/local/transforms.conf
[ta_builder_internal_use_table_format_results_for_badge]
DELIMS = ","
FIELDS = field_0,employee,door,result
splunk@show-demo-i-0a2741c4943a40369:~$
```

transforms.conf

More information about transforms.conf can be found in the Admin Manual:
https://docs.splunk.com/Documentation/Splunk/latest/Admin/Transformsconf

Map Source Type to the Authentication Data Model

12. Open the **Splunk Add-on Builder** and click on **Map to Data Models** in the menu bar.
13. Click on **New Data Model Mapping**.



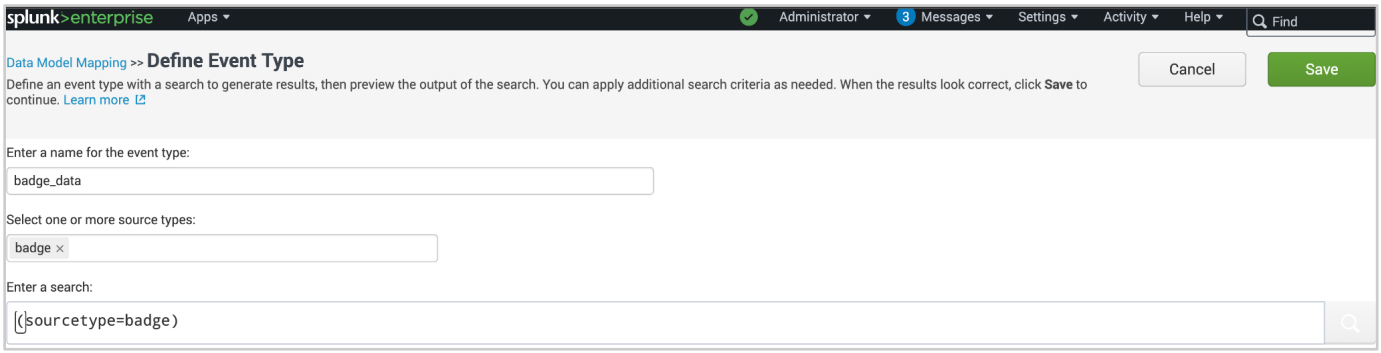
14. On the **Define Event Type** page, enter the following into the fields:

    Enter a name for the event type: **badge_data**

    Select one or more source types: **badge**

    Enter a search: `(sourcetype=badge)`

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Click on **Save**.

Optional Step - view the underlying .conf files that get built as we adjust the add-on in the GUI

15. Verify the **eventtypes.conf** file by running the following CLI command to list the files in the local directory of your TA:

```
ls /opt/splunk/etc/apps/<TA_FOLDER_NAME>/local
```

You should see the eventtypes.conf file listed:

```
splunk@show-demo-i-07ab6aba7a3b60f63:/opt/splunk/etc/apps/TA-badge/local$ ls
app.conf  eventtypes.conf  props.conf  tags.conf  transforms.conf
```

16. View the contents of **eventtypes.conf** and see the newly defined **[badge_data]** stanza:

```
sudo cat /opt/splunk/etc/apps/<TA_FOLDER_NAME>/local/eventtypes.conf
```

You should see the **[badge_data]** stanza and the accompanying search you defined:

```
splunk@show-demo-i-07ab6aba7a3b60f63:/opt/splunk/etc/apps/TA-badge/local$ sudo cat eventtypes.conf
[badge_data]
search = (sourcetype=badge)
splunk@show-demo-i-07ab6aba7a3b60f63:/opt/splunk/etc/apps/TA-badge/local$
```

---

eventtypes.conf

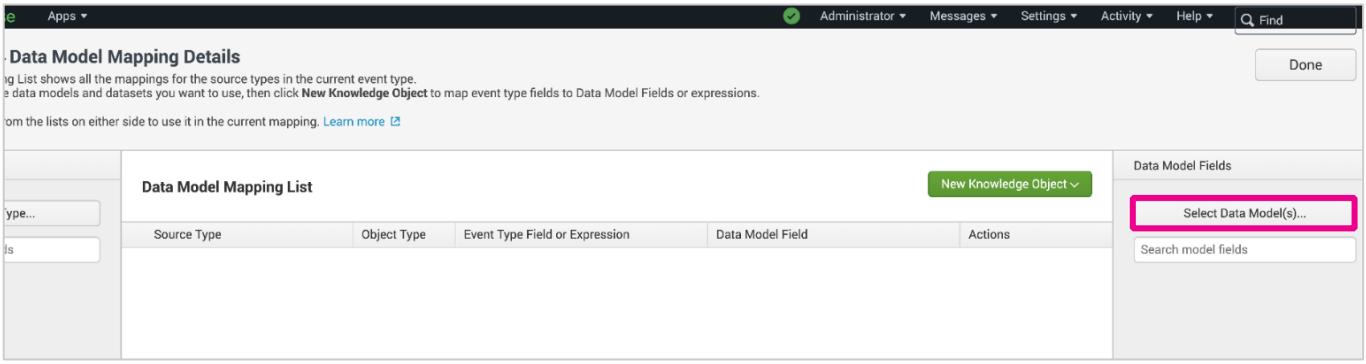
More information about eventtypes.conf can be found in the Admin Manual:
https://docs.splunk.com/Documentation/Splunk/latest/Admin/eventtypesconf

---

17. We will now choose which data model to map to.

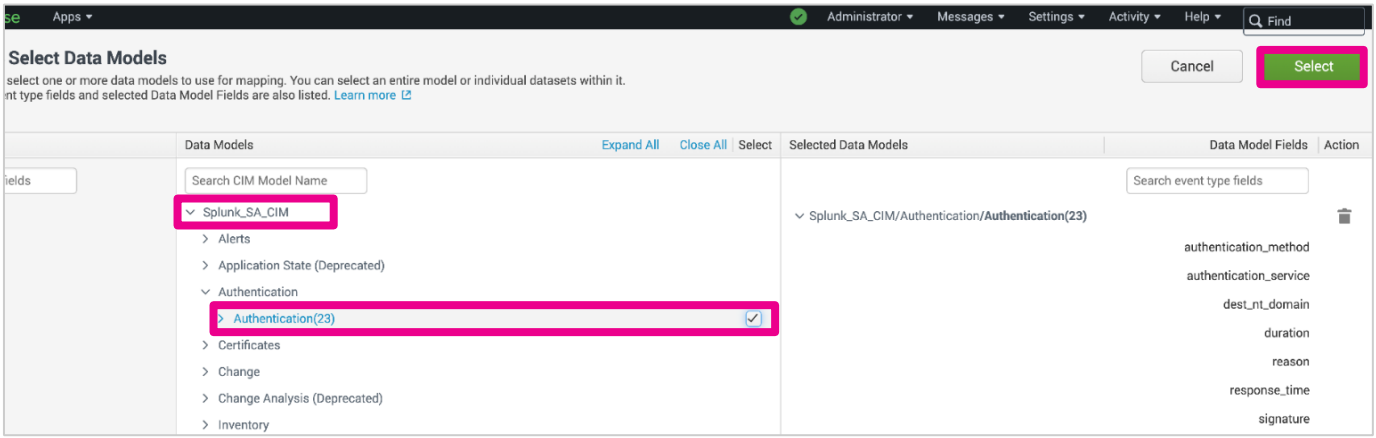
    Back in the Splunk GUI, you should be on the **Data Model Mapping Details** page, where you can add knowledge objects to enhance the badge data. On the right side of the page click on **Select Data Model(s)...**

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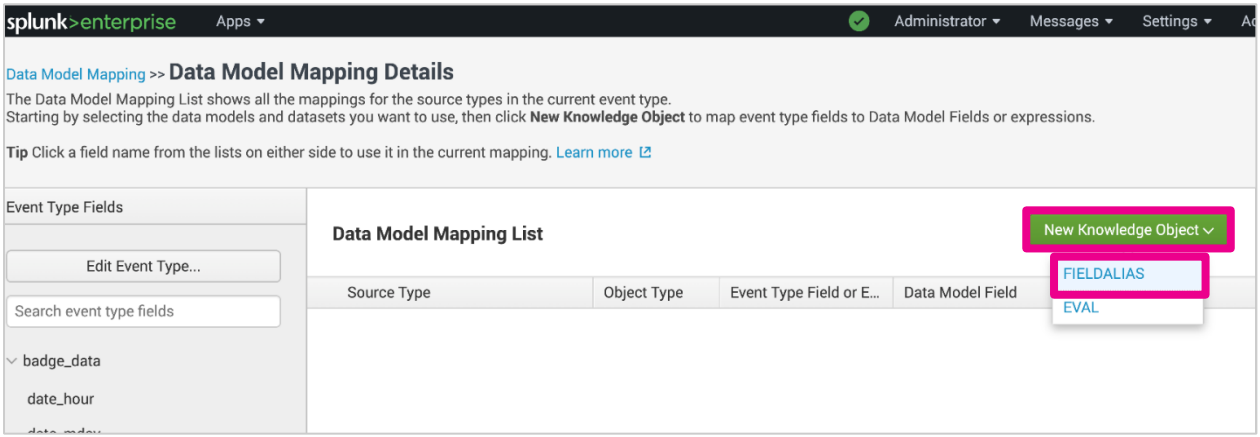


18. Under the **Data Models** panel, expand out the **Splunk_SA_CIM** list of data models. Select the **Authentication** data model and the corresponding Data Model Fields will appear on the right side of the page.

    Click on **Select**.



19. On the **Data Model Mapping Details** page, add the field alias knowledge object by clicking on **New Knowledge Object > FIELDALIAS**.



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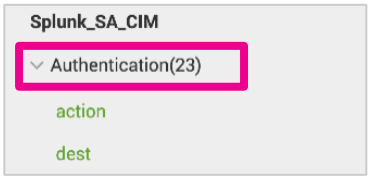
ℹ️ Field Aliases

Field aliases are an alternate name that you assign to a field, allowing you to search for those events using both the original field name and/or the alias.

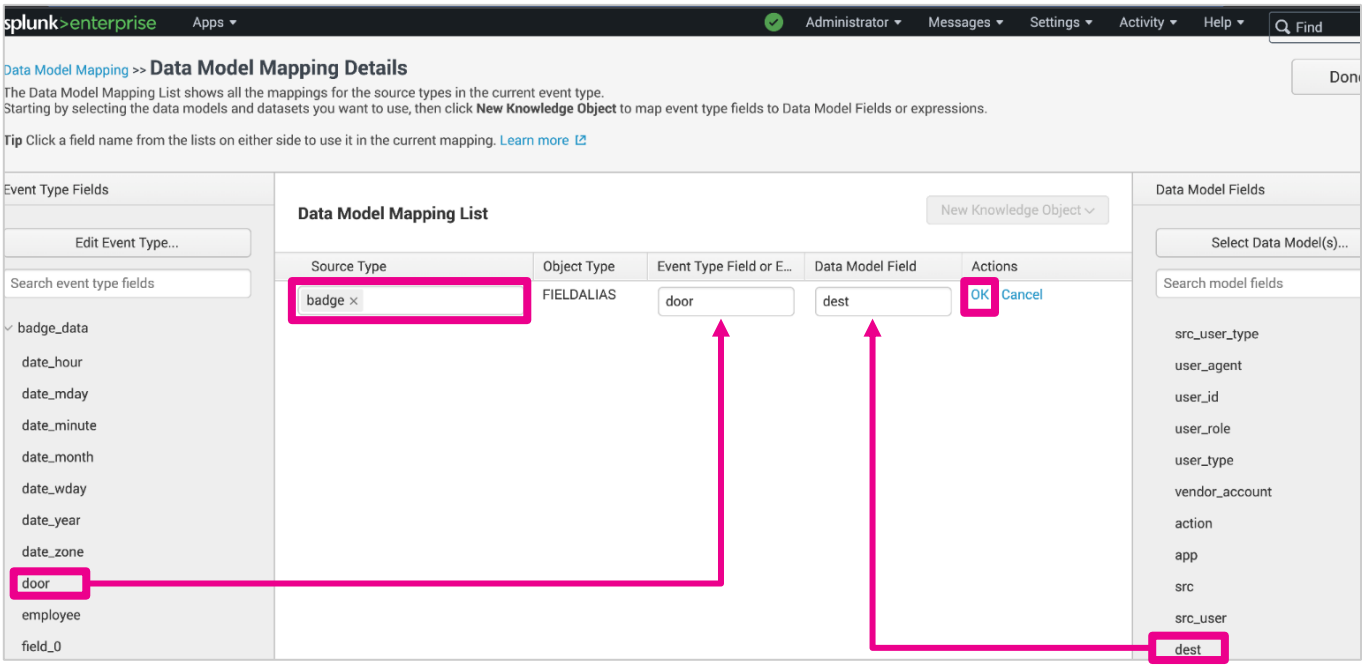
More information about field alias can be found in the Knowledge Manager Manual:
https://docs.splunk.com/Documentation/Splunk/latest/Knowledge/Addaliasestofields

20. On the next page in the **Data Model Mapping List** panel, select the **badge** source type.
21. In the **Event Type Fields** section on the left, expand out the **badge_data** section (if not already expanded) and click on "**door**" - this will populate the **Event Type Field or Expression** column with this field name.
22. In the **Data Model Fields** panel on the right, click on "**dest**" - this will populate the **Data Model Field** name.

    Note: If you do not see a list of fields in the **Data Model Fields** panel, you may need to expand the **Authentication(23)** section to view them.

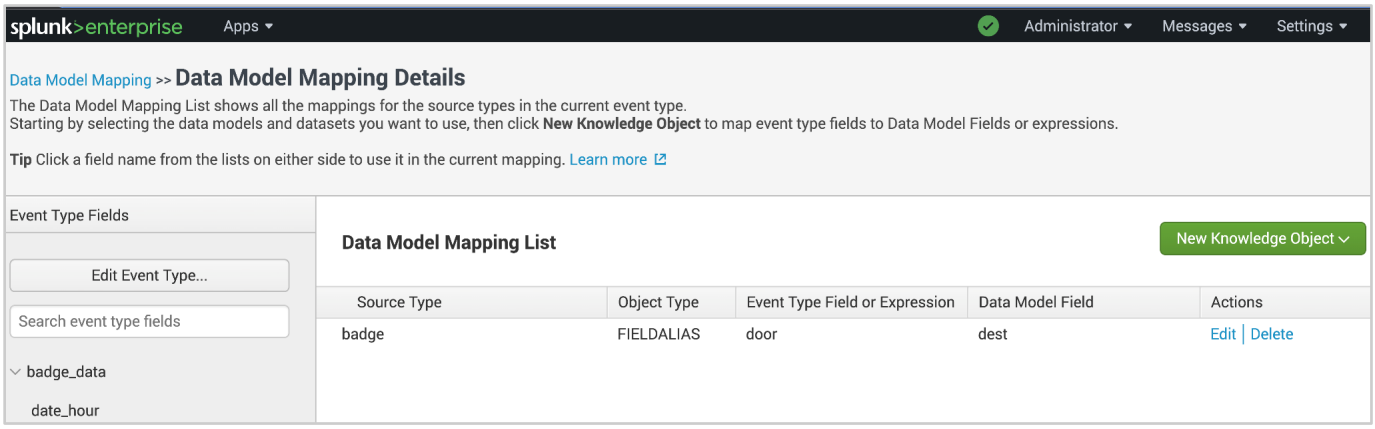


When the values are correctly populated, click on **OK** to save them.



23. Your mapping will now look like this:

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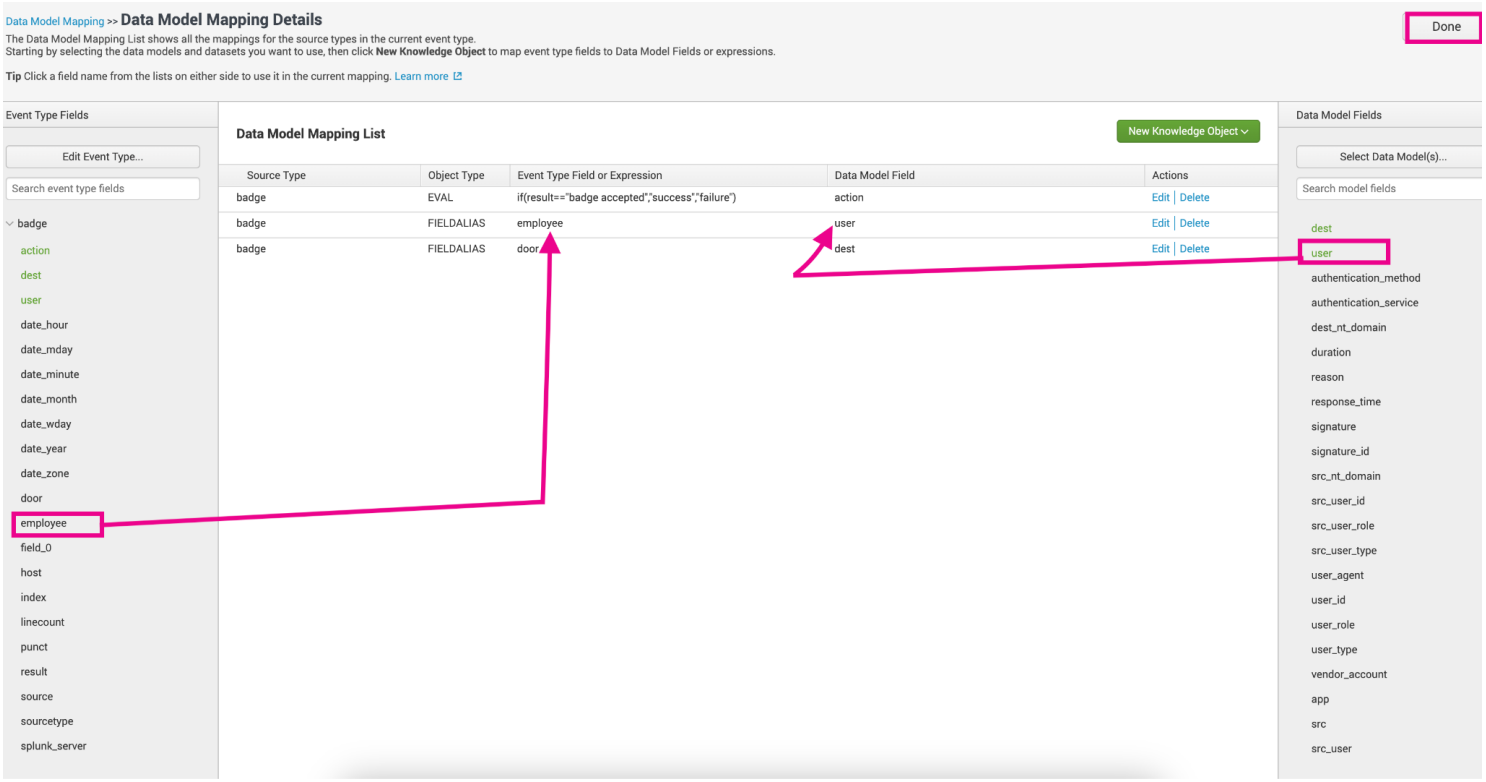
24. Now you will repeat step 19 to create a new field Alias for Employee

25. On the next page in the **Data Model Mapping List** panel, select the **badge** source type.

26. In the **Event Type Fields** section on the left, expand out the **badge_data** section (if not already expanded) and click on "**employee**" - this will populate the **Event Type Field or Expression** column with this field name.

27. In the **Data Model Fields** panel on the right, click on "**user**" - this will populate the **Data Model Field** name.

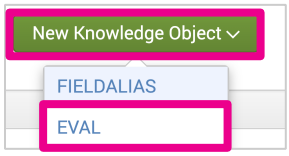
    Your Mapping will now look like this:



28. We will now add another knowledge object to populate the **action** field depending on whether a badge scan was successful or unsuccessful. To do this we will use an EVAL knowledge object (this works the same way as using `eval` in a search query).

Add the new knowledge object by clicking on **New Knowledge Object > EVAL**.

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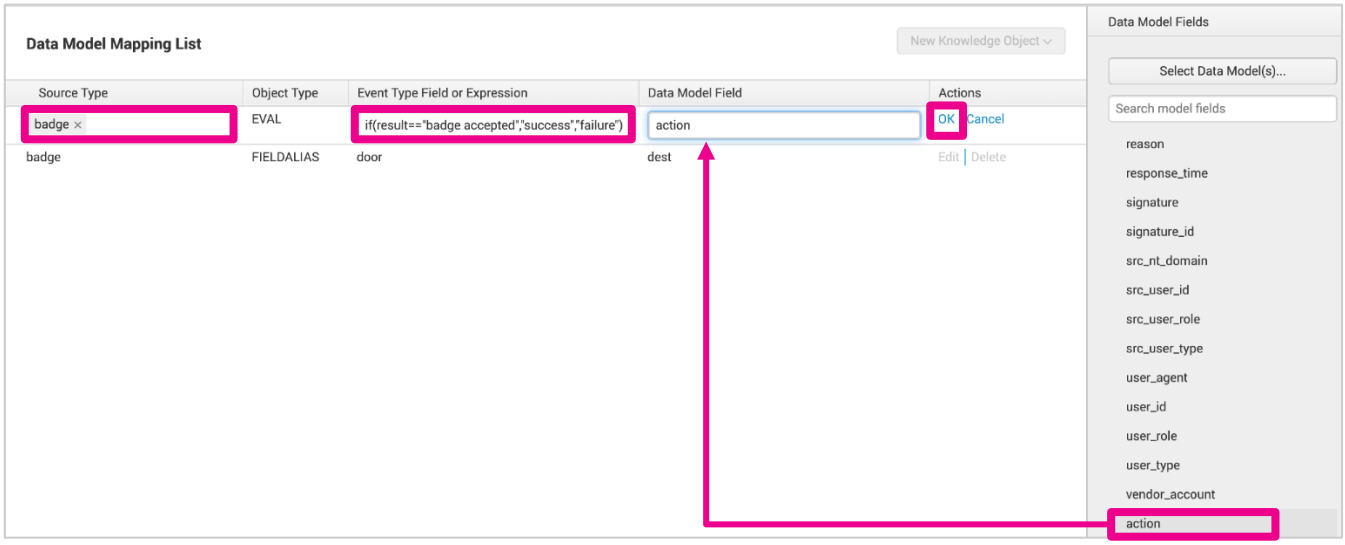
29. In the **Data Model Mapping List** panel, select the **badge** source type.

30. For **Event Type Field or Expression**, enter:

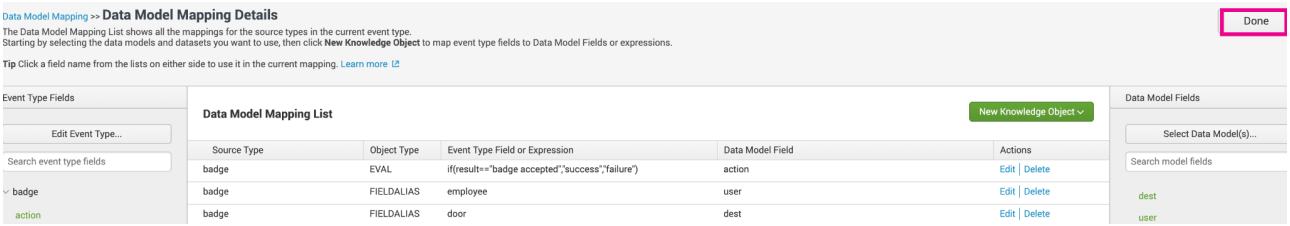
```
if(result=="badge accepted","success","failure")
```

31. For the **Data Model Field**, add the **action** field by clicking on it under the **Data Model Fields** panel on the right side of the page.

When the values are correctly populated, click on **OK** to save them.



32. When your complete mapping looks like this, click on **Done**.



Rebuild Your Data Model

---

33. Now that we have updated our data model mappings we need to rebuild it so Splunk regenerates the summaries for the data model. This will ensure that our search results are both fast and accurate.

To do this, navigate to **Settings > Data Models> Expand Authentication Data model > Choose Rebuild**

Wait for this to complete, this could take a few seconds to a number of minutes.

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Verify the Authentication Dashboard

---

34. Navigate to **Apps > DataOnboarding4Ninjas** and click on **Authentication Dashboard** in the menu bar.

The dashboard should now be populated. This is because the underlying dashboard searches are looking for events that map back to the Authentication data model - events that have fields like **action** and **dest**. While our original events may not have contained action and dest, we have now mapped our events to those data model field names and our data is now being returned in the results.



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Appendix

Troubleshooting for Lab 1, Option 2: Monitoring Through the CLI

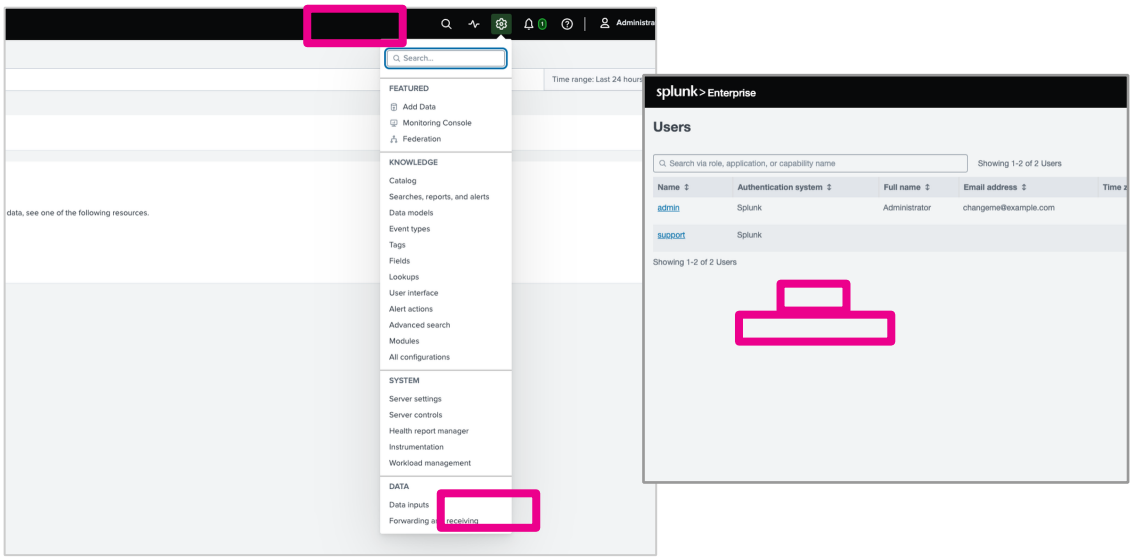
If an error was made in the monitoring stanza - for example, a typo or host_segment was set incorrectly - there are a couple of steps to fix the monitoring input.

1. Adjust the monitoring stanza in inputs.conf. To do this, edit the local inputs.conf file:

```
vi /opt/splunk/etc/system/local/inputs.conf
```

2. Type **" i "** to insert text into the file and adjust any errors in the monitoring stanza
3. Exit the vi editor by pressing the **escape** key then typing **:wq** and hitting **enter**. This is telling the editor to save and exit back to the main CLI.
4. Clean up the incorrectly parsed event data in Splunk using the `delete` command. This will prevent the incorrect event data from showing up in search results.

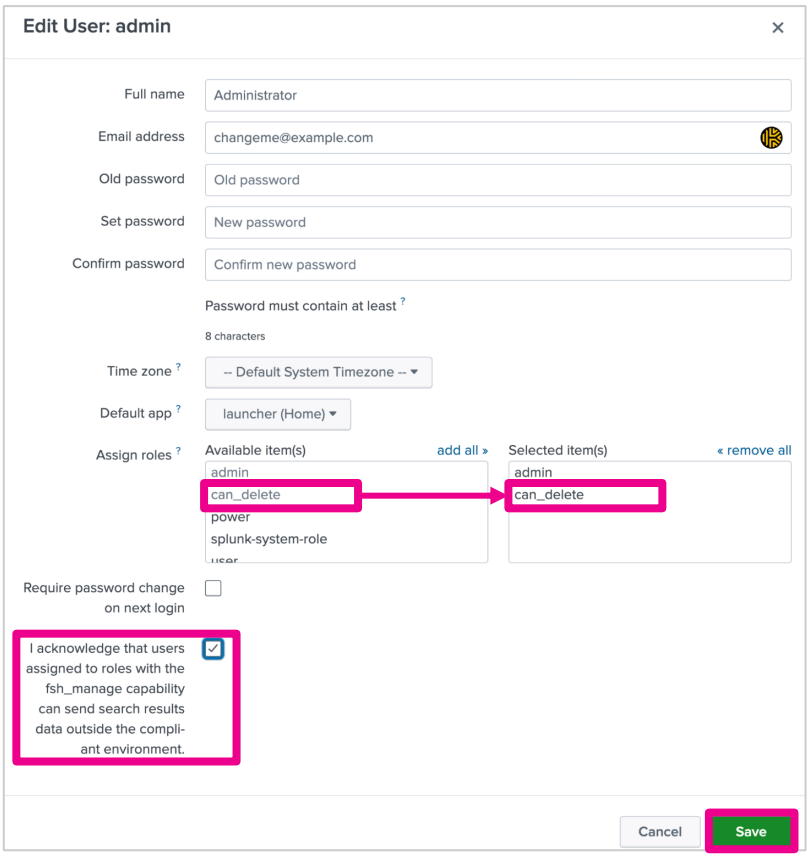
To do this, log in to Splunk and go to **Settings > Users and** click **Edit** next to the **admin** user. On the dropdown menu select **Edit**.



5. Under **Available item(s)** click on the **can_delete** role - this will add it to the **Selected item(s)** list on the right side of the screen.

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6. Check the "**I acknowledge...**" box and click on **Save**.



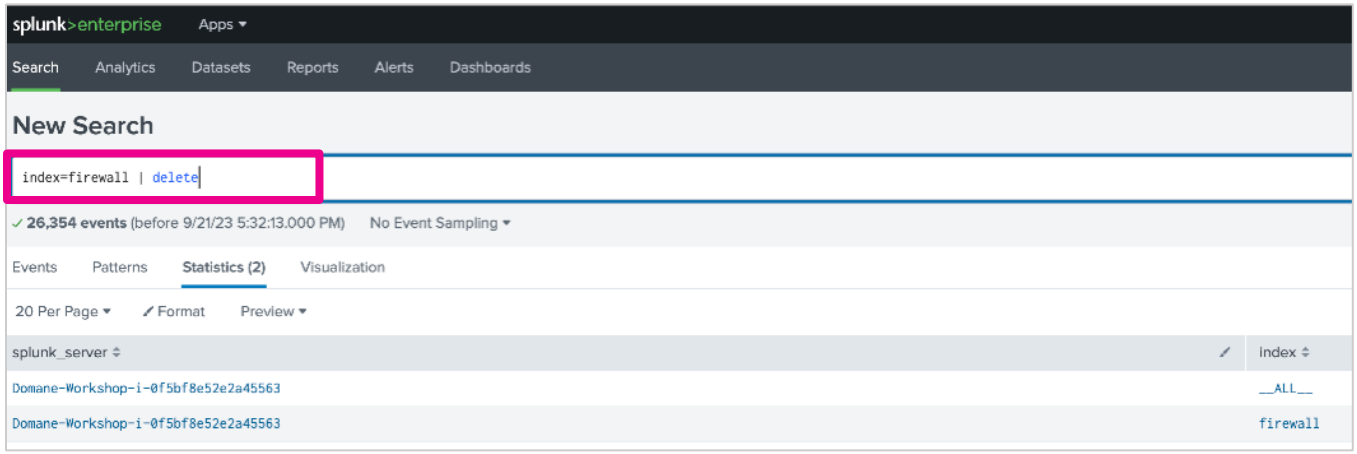
⚠ can_delete

The can_delete role is disabled by default for admin users to prevent accidental deletion of data. It is recommended to remove this role once you have deleted the required data.

---

7. Remove the incorrectly parsed event data by running the following search over **All time**. This will remove all of the previously ingested data in our **firewall** index:

```
index=firewall | delete
```



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Before we can re-ingest the firewall data (with the correct settings from inputs.conf) we need to clean the fishbucket - this will tell Splunk to clear out the cached markers for where it was monitoring our inputs and will ensure that Splunk will start collecting data from the very beginning of the log files.

To do this, open your terminal app and run the following commands:

8. Switch to the root user:

```
sudo su
```

9. Stop Splunk:

```
/opt/splunk/bin/splunk stop
```

10. Reset the fishbucket cache for each of the paths it was monitoring:

```
/opt/splunk/bin/splunk cmd btprobe -d /opt/splunk/var/lib/splunk/fishbucket/splunk_private_db/ --file
/opt/data/syslog/device1/firewall.log --reset
```

```
/opt/splunk/bin/splunk cmd btprobe -d /opt/splunk/var/lib/splunk/fishbucket/splunk_private_db/ --file
/opt/data/syslog/device2/firewall.log --reset
```

```
/opt/splunk/bin/splunk cmd btprobe -d /opt/splunk/var/lib/splunk/fishbucket/splunk_private_db/ --file
/opt/data/syslog/device3/firewall.log --reset
```

11. Start Splunk:

```
/opt/splunk/bin/splunk start
```

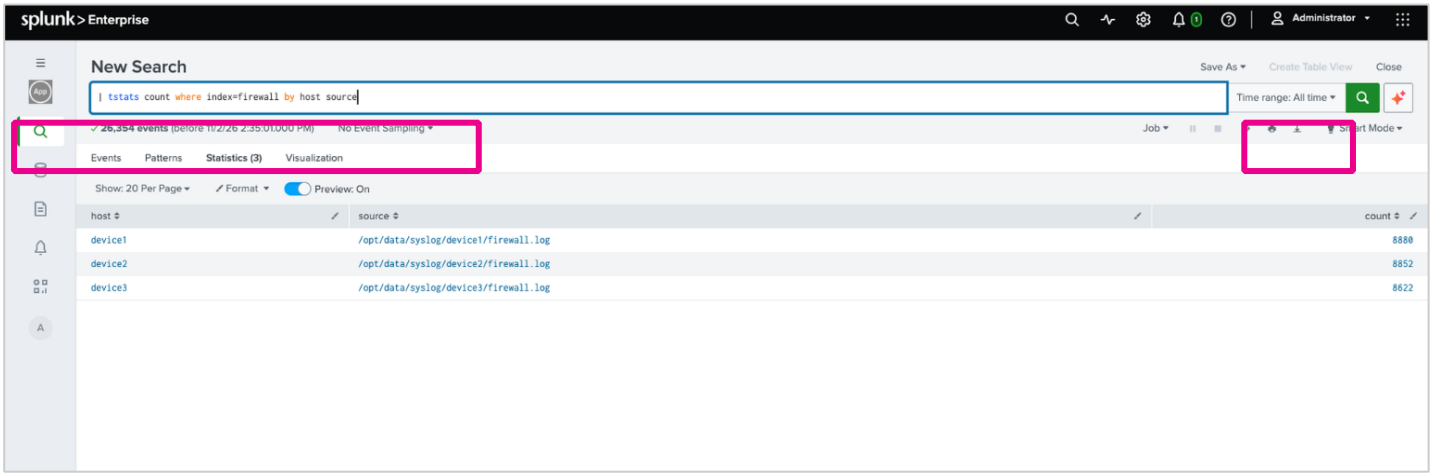
```
root@show-s4x-config-i-0f5bf8e52e2a45563:/opt/splunk/etc/system/local# /opt/splunk/bin/splunk stop
Stopping splunkd...
Shutting down.  Please wait, as this may take a few minutes.
...
root@show-s4x-config-i-0f5bf8e52e2a45563:/opt/splunk/etc/system/local# /opt/splunk/bin/splunk cmd btpr
obe -d /opt/splunk/var/lib/splunk/fishbucket/splunk_private_db/ --file /opt/data/syslog/device1/firewa
ll.log  --reset
Using logging configuration at /opt/splunk/etc/log-cmdline.cfg.
key=0x7fd9fc088298b571 scrc=0xbf57f14e210a29d5 sptr=5162816 fcrc=0xb84364badd1d748d flen=0 mdtm=154754
5184 wrtm=1695317097
Record (key 0x7fd9fc088298b571) reset.
root@show-s4x-config-i-0f5bf8e52e2a45563:/opt/splunk/etc/system/local# /opt/splunk/bin/splunk cmd btpr
obe -d /opt/splunk/var/lib/splunk/fishbucket/splunk_private_db/ --file /opt/data/syslog/device2/firewa
ll.log  --reset
Using logging configuration at /opt/splunk/etc/log-cmdline.cfg.
key=0xa6f86021bf0b84fc scrc=0x414ea3b3cc1a4916 sptr=5146581 fcrc=0xc3f60511ff64c816 flen=0 mdtm=154754
5184 wrtm=1695317097
Record (key 0xa6f86021bf0b84fc) reset.
root@show-s4x-config-i-0f5bf8e52e2a45563:/opt/splunk/etc/system/local# /opt/splunk/bin/splunk cmd btpr
obe -d /opt/splunk/var/lib/splunk/fishbucket/splunk_private_db/ --file /opt/data/syslog/device3/firewa
ll.log  --reset
Using logging configuration at /opt/splunk/etc/log-cmdline.cfg.
key=0xe9d1061cd785448f scrc=0x414ea3b3cc1a4916 sptr=5012859 fcrc=0x60b2c4eb45986862 flen=0 mdtm=154754
5184 wrtm=1695317097
Record (key 0xe9d1061cd785448f) reset.
root@show-s4x-config-i-0f5bf8e52e2a45563:/opt/splunk/etc/system/local# /opt/splunk/bin/splunk start
```

12. Validate your data by logging in to Splunk and running the following search over **All time**:

```
| tstats count where index=firewall by host source
```

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13. Results should be a table with 3 columns:



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Optional Activity: Lab 2, Tasks 1-4 Accomplished Through the CLI

Description:

These 2 tasks are optional replacements of tasks 1-4 in Lab 2. You will create a sourcetype called **badge** that correctly parses our badge.log data. Then you will bring in the badge.log data via a CLI command called oneshot. You will then assign that data to the sourcetype named **badge** and the index named **badge**. Finally, you will verify that parsed events are coming into the UI.

Establish a Custom “badge” Source Type

---

1. Open your terminal app and run the following commands:
2. Switch to the root user:

```
sudo su
```

3. Navigate to the DataOnboarding4Ninjas directory:

```
cd /opt/splunk/etc/apps/DataOnboarding4Ninjas/
```

4. Create and open a **props.conf** file using the **vi** visual editor:

```
vi props.conf
```

5. Press the “ **i** ” key to start editing the file:

```
i
```

6. Within the configuration file, type or paste in the following stanza:

```
[badge]
SHOULD_LINEMERGE=false
LINE_BREAKER=(##)
TIME_PREFIX=\d+\.\d+\.\d+\.\d+\s
TIME_FORMAT=%m%d%y %H:%M
MAX_TIMESTAMP_LOOKAHEAD=14
TRUNCATE=1000
```

7. Exit the vi editor by pressing the **escape** key then typing **:wq** and hitting **enter**. This is telling the editor to save and exit back to the main CLI.
8. Restart Splunk:

```
/opt/splunk/bin/splunk restart
```

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Use Oneshot to Ingest and View badge.log Data

9. Ingest data using oneshot by running the following command in the CLI. This is the same as “index once” in the Splunk GUI.

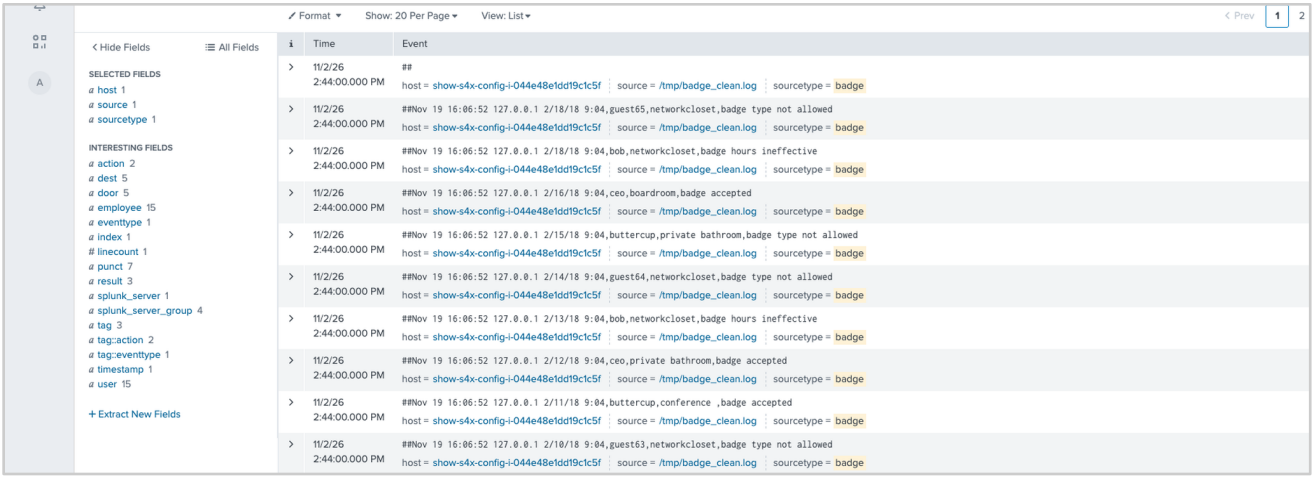
```
/opt/splunk/bin/splunk add oneshot /opt/data/badge.log -index badge -sourcetype badge
```

10. Verify the badge data in Splunk by logging in to Splunk and navigating to the **DataOnboarding4Ninjas** app.

11. Click **Search** in the menu bar and run the following search over **All time**.

```
index=firewall sourcetype=badge
```

You should now see your badge data in the search results:



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